
The Accuracy of Matched Sample Surveys: Evidence from the Cooperative Election Study

Jefferson Pruett¹, Gabriel Li², Jon Krosnick^{3,*}

¹Department of Economics, Stanford University, Stanford, CA, US

²Assistant Professor, Department of Communication Studies, Chapman University, Orange, CA, US

³Professor, Departments of Communication, Political Science, and Psychology, Stanford University, Stanford, CA, US

Abstract Nonprobability samples have grown in popularity among survey researchers, with proponents claiming that they can deliver accurate estimates at lower cost than probability samples. This growth has occurred despite accumulating evidence that nonprobability samples are generally less accurate. A notable exception appears in a small number of evaluations of YouGov, particularly a Pew study, which found YouGov to be more accurate than other nonprobability providers and more accurate than some probability sample estimates. Because YouGov relies on sample matching—an approach not used in most recent nonprobability evaluations—these findings raise the possibility that sample matching may yield unusually high accuracy within nonprobability designs.

This paper presents the largest evaluation of YouGov data to date, assessing accuracy across nine waves of the Cooperative Election Study from 2006 to 2022. Across 3,927 demographic estimates, 908 turnout and voting-method estimates, and 4,705 candidate-choice estimates, we find that the CES is no more accurate than nonprobability samples generally, and that its accuracy did not improve over the 17-year period. Excluding turnout, the CES exhibited an RMSE of 6.1%; including turnout, 7.9%—levels comparable to prior evaluations of nonprobability samples. To separate the contribution of sample matching from post-survey adjustment, we assess accuracy using unweighted CES data and newly constructed best-practice post-stratification weights. Applying YouGov’s proprietary weights substantially improves accuracy, whereas conventional demographic post-stratification does not, replicating previous findings that conventional demographic post-stratification alone does not significantly improve accuracy in nonprobability samples.

Before 2000, social science survey research relied primarily on probability samples. With the rise of the internet, nonprobability sampling reemerged. Some firms initially adapted probability-based designs to online administration, improving efficiency in data collection and processing. Others pursued a different model, offering faster and cheaper surveys based on nonprobability online panels. Over time, these providers claimed increasingly sophisticated methods and the ability to deliver survey estimates that were not only fast and inexpensive, but also accurate.

However, existing evidence consistently finds that nonprobability samples are less accurate than probability samples (see Cornesse et al. 2020 for a review of 25 studies) and in some cases, strikingly inaccurate (MacInnis et al. 2018; Yeager et al. 2011). Nonetheless, some scholars and survey firms argue that advanced nonprobability methods can yield accurate

estimates (Ansolabehere and Rivers 2013; Crampton 2007), raising the possibility that many prior evaluations do not reflect the performance of more sophisticated vendors (Baker et al. 2013).

One nonprobability approach has appeared particularly promising: sample matching. Popularized by YouGov (Rivers 2007), sample matching adapts techniques from causal inference, where matched control groups are constructed to estimate treatment effects (Stuart 2010). Applied to internet surveys, this approach synthesizes a matched sample from nonrandom online panels to approximate a hypothetical probability sample Rivers (2006). YouGov (2025) claims to be one of the most quoted pollsters in the world, and its matched sample surveys are widely used across political science, having been cited over 4,200 times.¹

The present study is a large-scale evaluation of nonprobability matched-sample surveys. Only a few studies have examined the accuracy of such designs (Ansolabehere and Rivers 2013; Ansolabehere and Schaffner 2014; Vavreck and Rivers 2008), and some evidence finds matching to perform no better than conventional post-stratification weighting or yields inconclusive results (Mercer, Lau, and Kennedy 2018; Brick et al. 2015; Buskirk and Dutwin 2015; DiSogra et al. 2015). Prior work has not separated the independent contributions of matching and weighting within the survey, nor assessed whether accuracy has improved over time. Existing evaluations also typically rely on a single survey year, an approach that can be misleading given the higher variance of nonprobability samples (MacInnis et al. 2018). Moreover, many assessments focus on offices included directly in the CES weighting scheme. Since 2014, for example, the CES has incorporated statewide U.S. Senate and gubernatorial outcomes into its post-stratification weights, mechanically constraining error for these offices and potentially conflating the effects of sample matching with those of weighting.

We address these issues using nine Cooperative Election Study (CES) surveys collected over 17 years. We first report accuracy using YouGov’s proprietary weights, focusing on variables not used in weight construction, and then evaluate accuracy without post-survey weighting and with newly constructed calibration weights, isolating the accuracy gains attributable to sample matching versus weighting. The scale and matched-sample design of the CES allow us to assess whether sample matching—and YouGov specifically—constitutes an outlier among nonprobability survey methods and vendors. Throughout, we distinguish primary variables, which were used to generate the CES weights and so are constrained to population levels, from secondary variables, which were not. This distinction is central to interpreting CES accuracy and may help explain the difference between prior valuations and ours.

¹Based on citations listed by Google Scholar, accessed January 25, 2025

Nonprobability Sampling and the Rise of YouGov

Probability sampling selects units at random with known inclusion probabilities, enabling valid population inference after appropriate weighting. In contrast, nonprobability samples rely on self-selected or otherwise uncontrolled recruitment, making inclusion probabilities unknowable and increasing the risk of selection bias, but are often simpler and less expensive to implement.

These inferential properties made probability sampling the foundation of major U.S. survey programs, including the Census Bureau’s Current Population Survey (CPS) and American Community Survey (ACS), the Bureau of Labor Statistics’ Consumer Expenditure Surveys, and the CDC’s National Health Interview Survey. After a series of high-profile polling failures in the 1930s and 1940s attributed to nonprobability designs—most notably the Gallup, Crossley, and Roper error in 1948 (Crossley 1937)—probability sampling became the standard for high-quality social scientific research (Cornesse et al. 2020).

The internet created new opportunities for survey data collection—and new commercial incentives. Early web-based survey providers initially relied on probability sampling, simply shifting questionnaires online to gain efficiency while preserving random selection. Firms soon recognized, however, that opt-in nonprobability panels offered even greater speed and cost savings. Although many argued that self-selected samples could not yield unbiased estimates and dismissed them as “quick and dirty” tools (Bethlehem 2010), their low per-respondent costs and rapid turnaround made web surveys attractive to researchers and organizations despite well-known sampling limitations (Wright 2017).

As nonprobability samples gained popularity, a substantial benchmarking literature emerged comparing their performance to high-quality probability samples. Across domains including voting behavior, consumption, sexual attitudes, and demographics, these studies generally found that nonprobability surveys were less accurate and more variable (Chang and Krosnick 2009; Dutwin and Buskirk 2017; Erens et al. 2014; MacInnis et al. 2018; Sturgis et al. 2018; Szolnoki and Hoffmann 2013; Yeager et al. 2011). Synthesizing evidence from 25 studies, Cornesse et al. (2020) concluded that “even in the age of declining response rates, accuracy in probability sample surveys is generally higher than in nonprobability sample surveys” (p. 22).

Researchers began to ask whether post-survey adjustment—especially weighting—could compensate for selection bias in opt-in panels and recover accurate population estimates. Evidence on this question remains mixed. A 2013 review found no consensus on whether such adjustments yield unbiased estimates (Baker et al. 2013), and subsequent work has been similarly divided. Some studies suggest that sophisticated methods can substantially

reduce bias: for example, Wang et al. (2015) showed that multilevel regression and post-stratification corrected a highly unrepresentative Xbox sample, while Dever (2018) found that propensity-score adjustments reduced bias in opt-in surveys.

Other studies have reached the opposite conclusion. Dutwin and Buskirk (2017) found that propensity weighting failed to reduce the roughly two-fold bias in nonprobability samples. Yeager et al. (2011) showed that although demographic weighting improved alignment with benchmark distributions, it did not improve substantive accuracy. Likewise, MacInnis et al. (2018) reported that demographic post-stratification consistently benefited probability samples but only inconsistently improved nonprobability samples, leaving their accuracy disadvantage largely intact. Taken together, this evidence highlights the uncertain and often limited impact of weighting in nonprobability surveys.

Despite these mixed findings, not all nonprobability survey designs rely on the same methods. Some vendors have attempted to address the limitations of opt-in panels by incorporating design-based features intended to more closely approximate probability samples. Among online nonprobability providers, YouGov has often been viewed as a distinctive case.

YouGov drew early attention after outperforming traditional telephone pollsters in several U.K. parliamentary elections, including the 2001 and 2005 contests (Coleman 2001; *Track Record: 2005 Errors* 2005). Following its 2007 acquisition of Polimetrix, YouGov entered the U.S. market and adopted Polimetrix’s matching-based methodology. Ahead of the 2008 presidential election, the firm emphasized its ability to deliver rapid, low-cost survey estimates while maintaining high accuracy, positioning sample matching as a potential solution to the limitations of opt-in panels (Crampton 2007; Bradshaw 2008).

These claims received notable support in a 2016 Pew Research Center evaluation. In a head-to-head comparison of eight nonprobability vendors and a probability sample fielded on the same questionnaire, YouGov not only outperformed all other opt-in samples but also, on average, exceeded the accuracy of Pew’s probability sample estimate (Kennedy et al. 2016). YouGov exhibited the lowest measurement error across 20 benchmark variables and performed especially well among Black and Hispanic respondents, populations for which survey error is typically higher (Hopkins et al. 2024). These findings raised the possibility that sample matching represents a substantive advance in nonprobability survey design.

Subsequent work has evaluated YouGov using election outcomes. Ansolabehere and Rivers (2013) compared CES vote intentions to state-level election returns from 2006 to 2010, reporting RMSEs between 3.4% and 4.0% across statewide offices. Similarly, Rivers and Bailey (2009) found that a CES subsample from 2008 outperformed contemporaneous probability and nonprobability surveys in estimating state-level presidential vote shares.

YouGov’s accuracy has not been uniformly high. A 2023 YouGov poll reported unusually

high rates of Holocaust denial and opposition to legal abortion among young adults, estimates that diverged sharply from those obtained using probability samples. A replication using a probability-based design attributed these discrepancies to a high prevalence of inattentive or “bogus” respondents in YouGov’s panel (Mercer, Kennedy, and Keeter 2024).

These mixed findings motivate a broader, systematic evaluation of YouGov’s accuracy. This study provides the most comprehensive assessment to date, using multiple election-year surveys administered by YouGov for the Cooperative Election Study.

The Cooperative Election Study

The CES² is a large-scale national survey designed to be representative of U.S. citizens aged 18 and older. It measures public opinion, demographics, and vote choice across federal and state elections, including President, U.S. Senate and House, Governor, Attorney General, Secretary of State, State Senator, and State Representative. The CES is administered by YouGov on behalf of a consortium of university and organizational research teams.

Fielded in every federal election year, the CES consists of two online waves: a pre-election survey conducted in October and a post-election survey administered after Election Day, with all pre-election respondents re-invited to the post-election wave.

CES Sample Matching Procedure

Sample matching is possible when a large but usually unrepresentative “reservoir” of respondents, typically an opt-in survey panel, is available (Cochran 1977). A random *target sample* is drawn from a sampling frame, and units from the reservoir are matched to units in the target sample based on shared characteristics. When the pool of respondents is sufficiently large and diverse and the number of auxiliary variables used to identify matches is large, the matched sample can have joint distributions of the matching variables similar to those of the target sample. Rivers (2006) suggests that the matched sample can thus effectively be treated as a random sample.

For the years of the CES we examined, panelists drawn from multiple opt-in providers were invited to complete the CES questionnaires. From the set of completed responses, a matching function selected respondents whose characteristics most closely approximated the target sample. These respondents constituted the *matched sample*, which was subsequently adjusted to better align with the target sample. Table 3 summarizes key terms in this process.

²Before 2020, the CES was known as the Cooperative Congressional Election Study (CCES).

Table 1: Number of States with CES Respondents and Average Respondents per State, by Office and Year

Year	President	U.S. Senate	U.S. House	Governor	Atty. Gen.	Sec. of State	State Sen.	State Rep.
Number of States with CES Respondents								
2006	*	32	50	36	28	24	42	45
2008	51	33	12	11	-	-	39	41
2010	*	36	13	37	30	26	41	44
2012	51	33	43	11	10	7	40	43
2014	*	34	43	35	31	26	42	45
2016	51	34	43	12	10	8	41	43
2018	*	33	43	36	29	25	40	46
2020	51	34	43	11	9	7	41	44
2022	*	34	44	36	28	23	41	46
Average Number of CES Respondents per State								
2006	*	602	488	582	503	380	406	405
2008	470	351	510	319	-	-	409	454
2010	*	785	915	808	840	677	703	733
2012	770	857	841	492	680	619	663	708
2014	*	508	739	774	735	622	618	653
2016	881	934	950	548	780	586	742	813
2018	*	847	829	810	860	709	810	827
2020	889	700	971	558	981	705	816	892
2022	*	861	868	876	1065	826	784	800

* Office not on ballot that year.

- CES did not measure vote choice for that office that year.

Note: For U.S. House, respondent counts are summed across congressional districts within each state before averaging.

Table 2: CES Sample Description, 2006 – 2022

Year	Sample Size	Probability Sample?	Response Rate	Field Dates
2006	36,421	No	not published	November 7 – November 21
2008	32,800	No	46.8%	November 4 – November 18
2010	55,400	No	40.4%	November 2 – November 16
2012	54,535	No	34.5%	November 6 – November 20
2014	56,200	No	23.4%	November 4 – November 18
2016	64,600	No	13.9%	November 9 – December 14
2018	60,000	No	30.0%	November 7 – December 3
2020	61,000	No	61.2%	November 8 – December 14
2022	60,000	No	47.6%	November 10 – December 15

Source: Cooperative Election Study Common Content documentation for each year.

Note: All samples are non-probability samples recruited from online panels. Response rates calculated as completed interviews divided by initial invitations.

Table 3: CES Sample Matching Terms

Term	Definition
Population of Interest	All U.S. citizens aged 18 and older. The population about which the CES seeks to draw conclusions.
Target Sample	A stratified random sample drawn from the ACS. Serves as the hypothetical probability sample that the matched sample attempts to replicate. Exists only as records, not CES respondents.
Panel Reservoir	The full set of panelists from YouGov and partner providers who have agreed to take surveys. A large, nonprobability pool from which respondents may be recruited.
Invited Respondents	Panelists selected for invitation to the pre-election questionnaire, typically chosen using a cross-classification on demographic variables. Not all invited panelists respond.
Participating Respondents	Panelists who complete the pre-election (and, when applicable, post-election) questionnaire. These are the realized respondents from whom matches can be drawn.
Matched Sample	The subset of participating respondents whose characteristics most closely match the target sample according to YouGov’s matching function. This constitutes the analytic CES sample.
Weighted Matched Sample	The matched sample after post-survey weighting (propensity weighting or entropy balancing), intended to reduce remaining imbalance between the matched and target samples.

Enumeration of the Target Sample

The CES population of interest consisted of U.S. citizens aged 18 and older. Each year, the target sample was drawn from the prior year’s American Community Survey (ACS) using stratified random sampling. The population was partitioned into mutually exclusive and exhaustive strata (Table 4, and individuals were sampled with replacement within strata using ACS person weights. To achieve political as well as demographic representativeness, YouGov augmented the ACS frame with modeled political and religious variables derived from sources including the CPS Voting Supplement, the National Election Pool, the Pew Religious Landscape Survey, and prior CES waves, though specific modeling procedures were not publicly documented.

Sourcing Panel Respondents

YouGov maintains a large opt-in respondent panel recruited through online advertising and targeted outreach to improve demographic coverage. Panelists join by completing a registration survey that collects demographic and political information and are subsequently invited to participate in surveys in exchange for incentives (Kellner 2004).

Table 4: Strata Used to Construct CES Target Sample

Year	Strata
2006	Registration Status x State Size x District Competitiveness
2008	Age x Race x Gender x Education
2010	Age x Race x Gender x Education
2012	Age x Race x Gender x Education
2014	Age x Race x Gender x Education x Voter Registration
2016	Age x Race x Gender x Education x Voter Registration
2018	Not published
2020	Not published
2022	Not published

Source: Publicly-available CES data guides (<https://cces.gov.harvard.edu/>), published alongside primary datasets and which detail year-specific processes for sampling and weighting.

For the CES, matches to the target sample were drawn from both YouGov’s panel and external partner panels. In 2016, for example, the matched sample was selected from 106,382 completed questionnaires, roughly half of which came from YouGov panelists, with the remainder recruited through partner providers, listed in Table 5.

Table 5: Non-YouGov Panelist Providers

Year	External Providers
2006	None
2008	E-Rewards, Western Wats
2010	E-Rewards, Western Wats
2012	E-Rewards, Western Wats
2014	MyPoints, Research Now, Survey Sampling International
2016	MyPoints, Research Now, Survey Sampling International, Global Market Insights
2018	Dynata, Prodege, Critical Mix
2020	Dynata, Prodege, Critical Mix
2022	Dynata, Prodege, Disqo, Generation Lab

Source: <https://cces.gov.harvard.edu/> CES data guides

Matching Function

Matching for the CES began by selecting panelists from YouGov and partner panels using a six-way cross-classification of age, gender, race, education, region, and sample source. To account for nonresponse, multiple potential matches were assigned to each target unit based on estimated completion probabilities, with matching continuing until at least one

completed response was expected per target unit (Rivers 2006). From the resulting pool of completed questionnaires—which exceeded the desired sample size—the final matched sample was constructed by selecting the N respondents whose characteristics most closely aligned with the N units in the target sample (Rivers 2007). Matching was implemented using a weighted Euclidean distance metric, rather than exact or propensity score-based matching (Stuart 2010).

Formally, similarity between a target case t and a respondent i was defined by a weighted sum of variable-specific distance functions. For each matching variable k , the distance function $d_k(t_k, i_k)$ measured similarity on attribute k , and overall distance was computed as

$$D(t, i) = \sum_{k=1}^K w_k d_k(t_k, i_k),$$

where w_k denotes the relative importance of variable k in the matching procedure. Larger weights placed greater emphasis on close or exact matches for that attribute. For unordered categorical variables, YouGov employed predefined distance matrices specifying distances between category pairs. Although the specific variables varied slightly by year, matching generally incorporated gender, age, race, education, political interest, employment status, church attendance, and party identification.

Weighting

The matched sample was further weighted to address residual imbalance. From 2006 to 2012, the CES employed propensity score weighting, estimating a logistic model of inclusion in the sampling frame using age, education, gender, and turnout. Respondents were grouped into propensity score deciles, and the matched sample was post-stratified on these deciles, with additional post-stratification on gender, race, education, and age. Weights exceeding 7 were trimmed, and the final weights were normalized to the sample size.

From 2014 to 2022, the CES weighted the matched sample to the target sample using entropy balancing (Hainmueller 2012). This multivariate reweighting method assigns a scalar weight to each matched unit so that the reweighted sample satisfies specified covariate balance constraints, matching selected moments of the target sample and minimizing deviation from uniform weights (see Appendix .1).

The covariates used for entropy balancing varied by year but generally included gender, age, race, Hispanic origin, education, and recent presidential vote choice and turnout. The resulting weights were then further post-stratified on a subset of demographic and political variables. Beginning in 2014, weights exceeding 15 were trimmed, and final weights were normalized to the sample size. Table 6 lists the post-stratification variables used in each

year.

Table 6: Variables Used for Final Post-Stratification after Balancing

	2008	2010	2012	2014	2016	2018	2020	2022
Sex	✓	✓	✓	✓	✓	✓	✓	✓
Race	✓	✓	✓	✓	✓	✓	✓	✓
Education	✓	✓	✓	✓	✓	✓	✓	✓
Age	✓	✓	✓	✓	✓	✓	✓	✓
Turnout	✓	✓	✓					
Voter registration status				✓	✓	✓	✓	✓
State of Residence				✓	✓	✓	✓	✓
Vote choice for U.S. Senator				✓	✓	✓	✓	✓
Vote choice for Governor				✓	✓	✓	✓	✓
Of Hispanic origin						✓	✓	✓
“Born again” status						✓	✓	✓
2016 Presidential vote choice						✓	✓	✓
2020 Presidential vote choice							✓	✓

Note: The 2006 CES did not list any post-stratification variables in publicly-released information.

Benchmarks

Demographics

Population targets were drawn primarily from the Current Population Survey (CPS) (Flood et al. 2024), a probability survey widely used as a benchmark for U.S. population statistics (Hur and Achen 2013). CPS estimates are weighted to account for sampling design, nonresponse, and undercoverage, using population controls derived from the decennial census and updated annually. The interviewing process starts in person, and can be transitioned to telephone after a few months of interviewing. The CPS uses proxy reporting, where one person speaks on behalf of the whole household.

Election Returns and Turnout

Comprehensive state and federal election returns were compiled for all 50 states and the District of Columbia from 2006 to 2022. Presidential returns were obtained from the MIT Election Lab (MIT Election Data and Science Lab 2017), congressional and senatorial returns from the Federal Election Commission, gubernatorial returns from CQ Press, and

state-level offices from Ballotpedia. Voter turnout data were drawn from the University of Florida Election Lab (McDonald 2025). Any remaining results were obtained through direct correspondence with state election offices.

Methods

Measuring Error

CES error was evaluated by comparing survey estimates to external benchmarks. For vote choice, respondents reported whether they voted and, if so, for whom, under the assumption of accurate recall. Error was computed by comparing the CES-estimated vote share for the modal candidate—the candidate receiving the largest share of votes in the actual election—to that candidate’s true vote share. For the races where only party vote choice was available in the CES, error was computed using the modal party share. Error for other benchmarks, including demographics, turnout, and voting method, was assessed analogously using the modal category of each benchmark distribution, with the state as the unit of comparison.

Error is reported using the root mean squared error (RMSE), defined here as

$$\text{RMSE}_{v,w} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{P}_{\text{CES},i,w} - P_{\text{true},i})^2},$$

where v indexes the benchmark variable being evaluated (e.g., the share of college graduates, Donald Trump’s vote share, etc.); $\hat{P}_{\text{CES},i,w}$ is the CES estimate for a variable v in state (or, for congressional races, district) i under weighting scheme w ; $P_{\text{true},i}$ is the corresponding benchmark value; and N is the number of distinct benchmark comparisons included in the calculation (i.e., the number of errors). Table 7 lists the number of error observations for each variable and year.

RMSE penalizes large deviations more heavily than small ones, making it a useful summary measure of survey accuracy: surveys that are consistently close to benchmark values across states have lower RMSE than those with similar average accuracy but occasional large errors. Following MacInnis et al. (2018) and Yeager et al. (2011), we report error separately for variables used to construct survey weights, since these are explicitly constrained to match population benchmarks. Accordingly, we distinguish between *primary* variables—those used by the CES for post-stratification—and *secondary* variables, which were not. Table 6 lists post-stratification variables by year. Beginning in 2014, outcomes for President, U.S. Senate, and Governor were included in the weighting process, limiting the extent to which error in these contests reflects unconstrained survey performance.

Table 7: Number of Error Observations by Variable and Year

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022	Sum
<i>Demographic</i>										
Age Group	51	51	51	51	51	51	51	51	51	459
Sex	51	51	51	51	51	51	51	51	51	459
Hispanic Origin	-	-	51	51	51	51	51	51	51	357
Education	51	51	51	51	51	51	51	51	51	459
Family Income	51	51	51	51	51	51	51	51	51	459
Employment Status	51	51	51	51	51	51	51	51	51	459
Union Membership	51	51	51	51	51	51	51	51	51	459
Veteran Status	51	51	51	51	51	51	51	51	51	459
Residence Duration	-	51	51	51	51	51	-	51	51	357
<i>Voting Administration</i>										
Voting Turnout	50	50	50	50	50	50	50	50	50	450
Voting Method	50	51	51	51	51	51	51	51	51	458
<i>Candidate Choice</i>										
President	*	51	*	51	*	51	*	51	*	204
U.S. Senate	32	33	36	33	34	34	33	34	34	303
U.S. House	50	132	152	426	422	426	424	427	429	2,888
Governor	36	11	37	11	35	12	36	11	36	225
Attorney General	28	-	30	10	31	10	29	9	28	175
Secretary of State	24	-	26	7	26	8	25	7	23	146
State Senator	42	39	41	40	42	41	40	41	41	367
State Representative	45	41	44	43	45	43	46	44	46	397
Total Errors per Year	714	816	926	1,181	1,195	1,185	1,142	1,184	1,197	9,540

* Office not on ballot that year.

- CES did not measure that variable that year.

Note: Shows number of state-level observations (errors) for each variable by year. Washington D.C. is counted as the 51st state.

With the exception of U.S. House elections, the unit of analysis for all races is the state. For U.S. House contests, respondent congressional district was collected, so error could be additionally computed at the district level. We assess the relationship between per-district sample size and error separately.

Unlike prior CES evaluations that measured accuracy using Democratic vote share, we use candidate vote share where possible. As the CES evolved, the wording of vote-choice questions changed across years, with early waves asking respondents which *party* they voted for and later years asking them to select the *candidate* they voted for, with party labels provided alongside candidate names.

Collapsing vote choice to the party level can introduce mismeasurement. In reviewing election return data, we identified numerous cases in which party-level reporting obscured accuracy, including elections with multiple candidates sharing a party label, candidates appearing on multiple party lines, ballots absent party labels, or candidates switching party affiliation between the primary and general election. To minimize these issues, we measured error using candidate vote share whenever possible, matching CES responses to election returns and relying on party-level comparisons only when candidate-level information was unavailable. Candidate vote choice was available for presidential contests; for U.S. Senate and House elections beginning in 2008; for gubernatorial elections beginning in 2010; and for attorney general and secretary of state races beginning in 2018. State legislative contests were evaluated using party vote share in all years.

Weighting Scenarios

Comparing CES accuracy across weighting scenarios allows us to separate the contributions of sample matching and post-survey adjustment. Because sample matching is intended to approximate a random sample, error in the unweighted CES reflects the performance of matching alone, while differences between weighted and unweighted estimates capture the incremental accuracy provided by weighting.

CES weights reflect design choices specific to YouGov, including the incorporation of known U.S. Senate and gubernatorial outcomes. To assess whether conventional demographic post-stratification improves accuracy in nonprobability matched samples independent of these choices, we evaluated accuracy under an alternative weighting scheme applied consistently across years.

To this end, we generated an additional set of weights designed to minimize divergence between the CES and a set of demographic and turnout benchmarks drawn from the CPS and administrative sources. Only CES variables with strictly comparable CPS counter-

parts—identical or fully collapsible response categories measuring the same underlying construct—were eligible for inclusion in the weighting process. Notably, race was excluded because CPS race categories could not be exhaustively mapped to CES categories. As a result, although 29 CES variables had conceptual CPS counterparts across years, only 9–11 per year were used for weight generation (Table 8).

Table 8: Variables Used during ANESRake Weight Generation

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022
Age Group	✓	✓	✓	✓	✓	✓	✓	✓	✓
Education	✓	✓	✓	✓	✓	✓	✓	✓	✓
Employment Status	✓	✓	✓	✓	✓	✓	✓	✓	✓
Family Income	✓	✓	✓	✓	✓	✓	✓	✓	✓
Hispanic Origin	–	–	✓	✓	✓	✓	✓	✓	✓
Sex	✓	✓	✓	✓	✓	✓	✓	✓	✓
Union Membership	✓	×	✓	✓	✓	✓	✓	✓	✓
Veteran Status	✓	✓	✓	✓	✓	✓	✓	✓	✓
Voting Turnout	✓	✓	✓	✓	✓	✓	✓	✓	✓
Voting Method	✓	✓	✓	✓	✓	✓	✓	✓	✓
Residence Duration	–	✓	✓	✓	✓	×	–	✓	✓

✓ Variable used as a target for ANESRake weighting.

× Variable provided to ANESRake but not selected by the algorithm.

– Variable not provided to ANESRake (benchmark unavailable for that year).

Weights were then constructed following the procedures in DeBell and Krosnick (2009) using anesrake (Pasek 2018). In order from the most to least discrepant category, categories whose weighted CES proportions differed from CPS benchmarks by at least 5% were updated sequentially. At each step, current weights were multiplied by the ratio of the population proportion to the weighted sample proportion. Weights were capped at 5 and re-centered to maintain a mean of 1. Iteration continued until no benchmark category exceeded the 5% tolerance or until convergence.

Results

We begin by assessing CES accuracy under YouGov’s weighting scheme and examining trends over time. We then evaluate the performance of sample matching in isolation using unweighted data, before comparing these results to accuracy under alternative post-stratification weights generated via anesrake.

Overall CES Survey Error

Table 9 reports overall CES accuracy across all available measures using YouGov’s post-stratification weights. Pooling RMSE estimates across the nine survey waves yields 9,540 individual error observations and an average annual RMSE of 7.9%, with values ranging from a high of 9.6% in 2006 to a low of 6.5% in 2020. Accuracy varied by variable class. Demographic variables exhibited the lowest average error (5.8%), followed by candidate-choice variables (6.7%), while voting-administration measures showed markedly higher error (21.1%), driven by turnout misestimation, which alone exhibited an RMSE of 37.1%. Excluding turnout, overall RMSE was 6.1%; including it, 7.9%.

Table 9: Matching + Post-Stratification RMSE by Variable and Year

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022	Mean	SD
<i>Demographic</i>											
Age Group	7.5	4.5†	5.3†	3.5†	4.1†	2.9†	3.4†	3.4†	3.0†	4.2	1.4
Sex	5.4	6.7†	4.9†	1.5†	2.2†	1.0†	2.9†	2.5†	3.0†	3.4	1.9
Hispanic Origin	-	-	7.7	8.7	8.0	8.3	8.7†	8.2†	3.4†	7.6	1.9
Education	13.0	3.2†	6.4†	3.5†	3.9†	5.7†	4.2†	4.6†	3.6†	5.3	3.1
Family Income	4.5	3.5	3.9	3.8	4.4	4.5	5.2	6.7	8.7	5.0	1.7
Employment Status	7.2	10.8	10.9	13.9	12.0	12.8	13.3	13.8	12.0	11.8	2.1
Union Membership	5.3	3.7	5.6	5.9	5.0	5.1	5.8	6.3	5.3	5.3	0.7
Veteran Status	4.8	4.6	5.1	4.3	4.5	4.0	3.0	3.1	3.2	4.1	0.8
Residence Duration	-	6.0	7.3	4.8	5.1	5.0	-	4.2	6.3	5.5	1.1
Demographic Avg.	6.8	5.4	6.4	5.5	5.5	5.5	5.8	5.9	5.4	5.8	1.6
<i>Voting Administration</i>											
Voting Turnout	43.8	26.0†	31.6†	38.7†	47.2	36.7	35.1	26.5	48.1	37.1	8.2
Voting Method	3.7	4.5	6.0	4.7	3.5	5.5	5.0	6.8	6.3	5.1	1.1
Voting Administration Avg.	23.8	15.2	18.8	21.7	25.3	21.1	20.1	16.6	27.2	21.1	4.7
<i>Candidate Choice</i>											
President	*	3.5	*	2.9	*	1.7	*	2.4†	*	2.6	0.8
U.S. Senate	5.9	5.1	2.0	2.3	3.1†	4.0†	4.7†	3.3†	4.8†	3.9	1.3
U.S. House	4.9	9.4	11.4	9.8	9.0	9.6	9.5	8.0	9.9	9.1	1.8
Governor	5.0	4.7	3.5	3.4	3.5†	4.0†	4.5†	4.3†	5.7†	4.3	0.8
Attorney General	13.3	-	11.8	15.1	8.0	4.9	4.8	3.5	7.6	8.6	4.3
Secretary of State	11.2	-	6.4	10.9	8.3	9.6	4.2	3.6	13.9	8.5	3.6
State Senator	10.7	5.5	9.7	7.4	13.6	8.0	4.5	6.4	8.8	8.3	2.8
State Representative	7.3	4.9	5.4	6.4	6.3	6.5	4.6	5.5	8.5	6.2	1.2
Candidate Choice Avg.	8.3	5.5	7.2	7.3	7.4	6.0	5.3	4.6	8.5	6.7	2.1
Yearly RMSE	9.6	6.7	8.1	8.0	8.4	7.4	7.3	6.5	9.0	7.9	1.0
Excluding Turnout	7.3	5.4	6.7	6.3	6.1	5.7	5.5	5.4	6.7	6.1	0.6

† indicates Primary variables (used in CES post-stratification weighting for that year).

* indicates office not on ballot that year.

- indicates CES did not measure vote choice for that office that year.

Note: Means and standard deviations are computed across years for each variable.

Within demographic measures, the CES performed worst relative to CPS benchmarks

for employment status (11.8%) and best for sex (3.4%). Error likewise differed across offices: lowest for presidential elections (2.6%) and highest for U.S. House contests (9.1%).

Elevated error for turnout was consistent with a large literature documenting systematic overreporting of electoral participation in surveys, often attributed to social desirability bias and recall error (DeBell, Krosnick, et al. 2020). Unless noted otherwise our substantive conclusions rely on turnout-excluded figures. More recent work, however, emphasizes that discrepancies between survey responses and administrative benchmarks also reflect limitations in record linkage and administrative data quality, rather than respondent misreporting alone (Berent, Krosnick, and Lupia 2016).

Demographic measures exhibited low cross-wave dispersion, with most standard deviations between one and two percentage points, indicating stable accuracy over time. In contrast, turnout displayed both the highest average error and the greatest variability ($SD = 8.2$). Candidate-choice accuracy is generally stable for high-salience offices such as President and Governor ($SD = 0.8$), but more variable for down-ballot races. Notably, U.S. House contests exhibited relatively high average error (9.1%) coupled with low cross-year dispersion ($SD = 1.8$). By contrast, U.S. Senate races featured lower average error (3.9%) but greater variability across years ($SD = 1.3$).

Distinguishing Primary and Secondary Error

Variables included in CES post-stratification (Table 6) exhibited lower error than variables that were not. As shown in Figure 1 and Table 10, average RMSE for *primary* variables—those used in post-stratification—was 3.7%, compared to 7.0% for *secondary* variables, yielding an average within-year gap of 3.2 percentage points (Figure 2).

Distribution of Errors

Prior work has shown that while average accuracy may differ only modestly between probability and non-probability samples, non-probability samples typically exhibit greater variation, with a non-trivial share of very large errors (Yeager et al. 2011; MacInnis et al. 2018; Lavrakas et al. 2022; Walker, Pettit, and Robinson 2009; Chang and Krosnick 2009).

Figure 3 illustrates this pattern in the CES. Although average error was moderate for many outcomes, the error distributions for secondary variables were strongly right-skewed across all three variable classes, indicating that a meaningful fraction of estimates deviate substantially from their benchmarks. Voting administration variables exhibited the greatest dispersion ($SD = 18.5$), but demographic ($SD = 4.8$) and candidate-choice variables ($SD = 6.1$) also displayed dispersions on par with previous findings for nonprobability samples

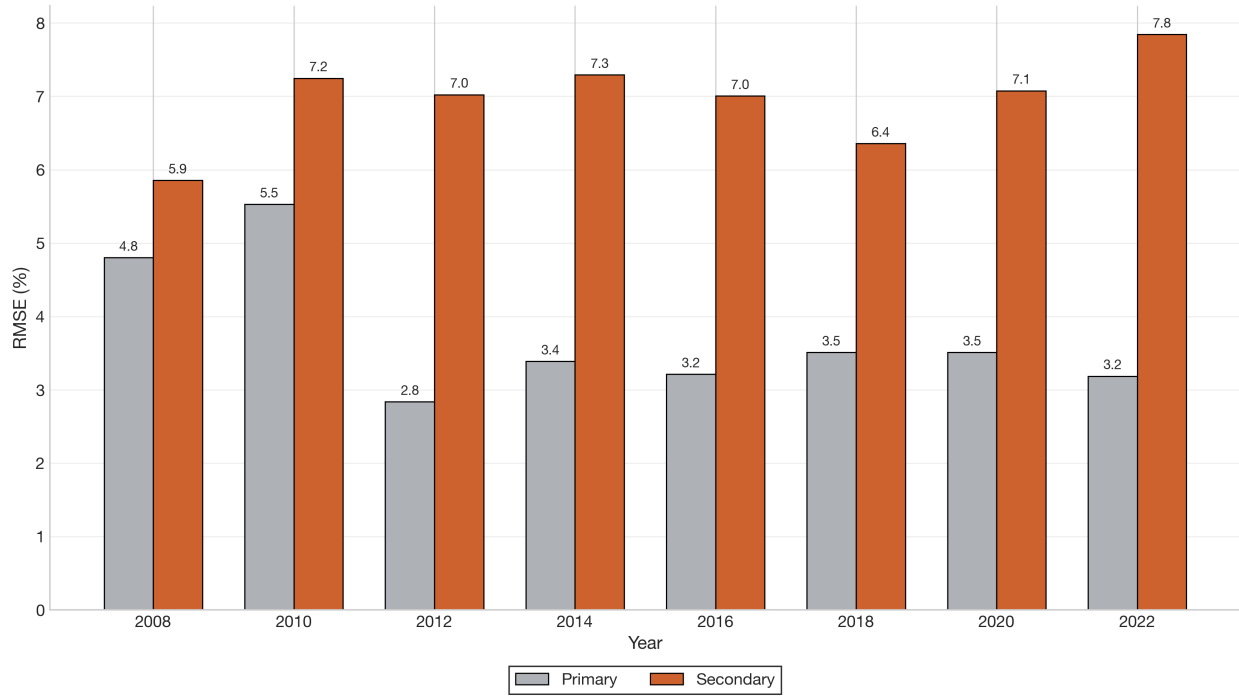


Figure 1: Primary vs. Secondary Variable RMSE

Primary variables (3): Age Group, Education, Sex. Secondary variables (8): Family Income, Employment Status, State Representative, State Senator, U.S. House, Union Membership, Veteran Status, and Voting Method. Only variables consistently classified as Primary or Secondary across all years and present in every CES wave are included. 2006 is excluded because the CES does not report post-stratification variables for that year.

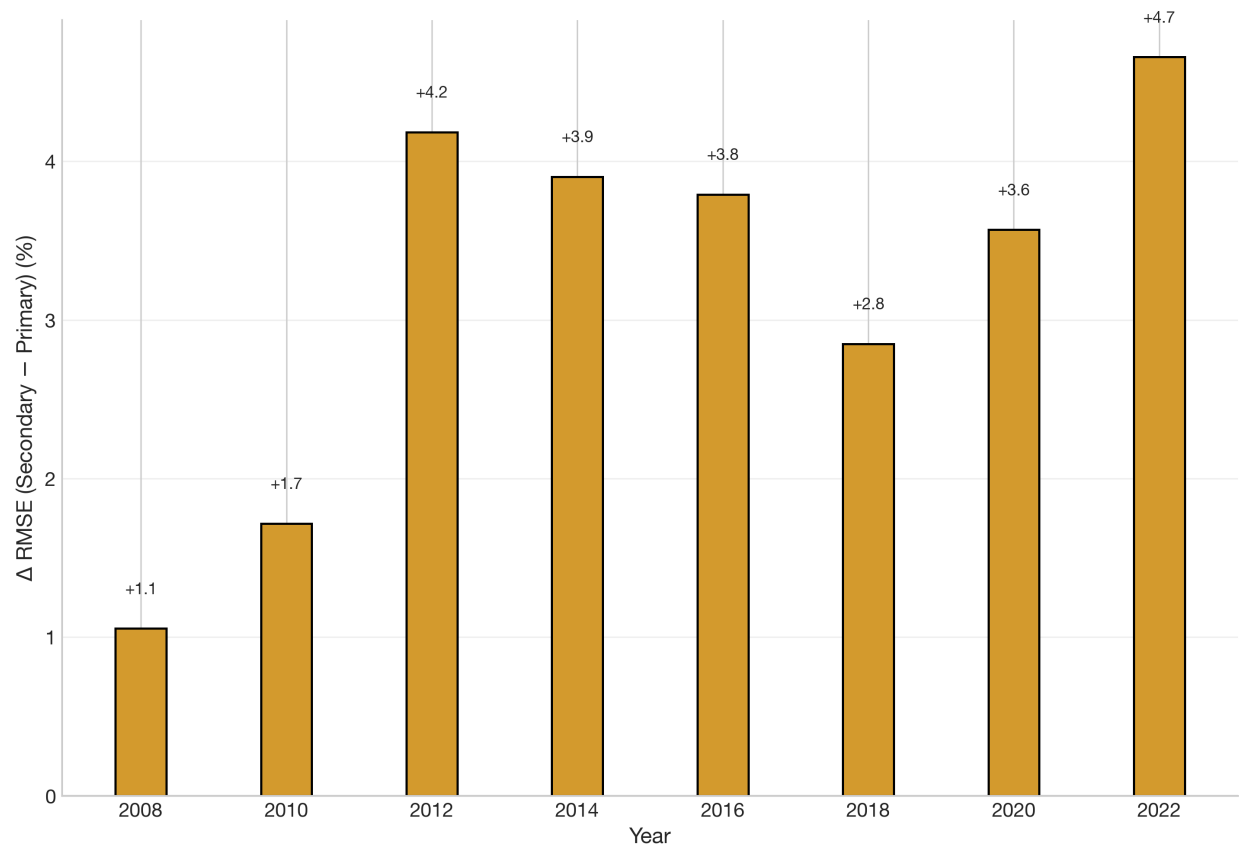


Figure 2: Difference between Primary and Secondary by Year

Positive values indicate Secondary variables had higher RMSE than Primary variables.

Table 10: Primary vs Secondary Variable RMSE by Year (2008–2022)

Variable	2008	2010	2012	2014	2016	2018	2020	2022	Mean	SD
<i>Primary Variables – Used to Construct CES Weights</i>										
Age Group	4.5	5.3	3.5	4.1	2.9	3.4	3.4	3.0	3.8	0.8
Education	3.2	6.4	3.5	3.9	5.7	4.2	4.6	3.6	4.4	1.1
Sex	6.7	4.9	1.5	2.2	1.0	2.9	2.5	3.0	3.1	1.7
Primary Avg.	4.8	5.5	2.8	3.4	3.2	3.5	3.5	3.2	3.7	0.9
<i>Secondary Variables – Not Used to Construct CES Weights</i>										
Employment Status	10.8	10.9	13.9	12.0	12.8	13.3	13.8	12.0	12.4	1.1
Family Income	3.5	3.9	3.8	4.4	4.5	5.2	6.7	8.7	5.1	1.7
State Representative	4.9	5.4	6.4	6.3	6.5	4.6	5.5	8.5	6.0	1.1
State Senator	5.5	9.7	7.4	13.6	8.0	4.5	6.4	8.8	8.0	2.6
U.S. House	9.4	11.4	9.8	9.0	9.6	9.5	8.0	9.9	9.6	0.9
Union Membership	3.7	5.6	5.9	5.0	5.1	5.8	6.3	5.3	5.3	0.7
Veteran Status	4.6	5.1	4.3	4.5	4.0	3.0	3.1	3.2	4.0	0.7
Voting Method	4.5	6.0	4.7	3.5	5.5	5.0	6.8	6.3	5.3	1.0
Secondary Avg.	5.9	7.2	7.0	7.3	7.0	6.4	7.1	7.8	7.0	0.6
Δ (Secondary – Primary)	+1.1	+1.7	+4.2	+3.9	+3.8	+2.8	+3.6	+4.7	+3.2	1.2

Note: 2006 is excluded because the CES does not report post-stratification variables for that year, making it impossible to classify variables consistently as primary or secondary. Variables listed are those that were consistently classified as primary or secondary in every CES wave from 2008 to 2022. Mean and SD are computed across years for each variable.

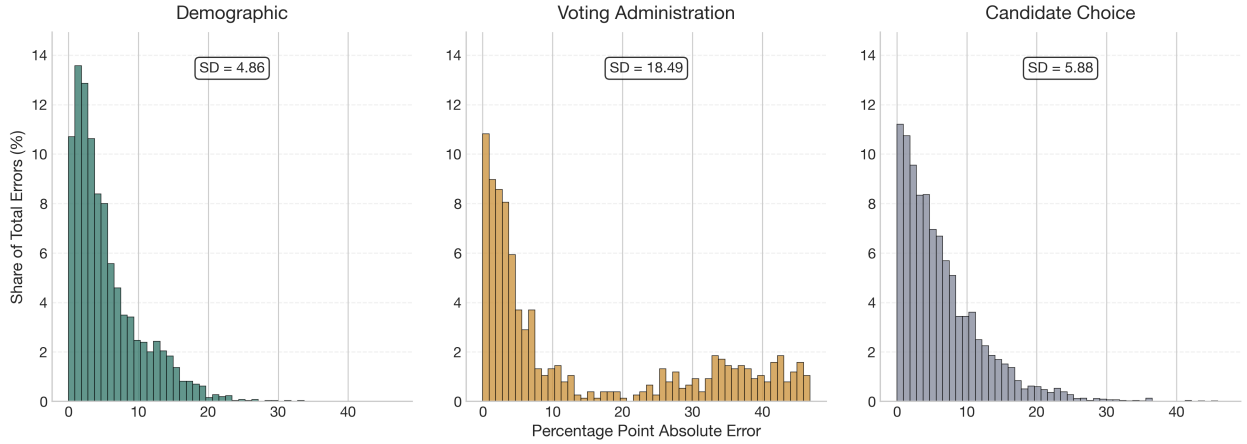


Figure 3: Error Distributions by Variable Class

Demographic variables (n=9): Age Group, Education, Employment Status, Family Income, Hispanic Origin, Residence Duration, Sex, Union Membership, and Veteran Status. Voting Administration variables (n=2): Voting Method and Voting Turnout. Candidate Choice variables (n=8): Attorney General, Governor, President, Secretary of State, State Representative, State Senator, U.S. House, and U.S. Senate. Each histogram shows the share of total errors at each level of percentage point absolute error. Turnout and voting method exhibit the widest error distribution (SD = 18.49), while Demographic variables show the narrowest (SD = 4.86)

(Yeager et al. 2011; MacInnis et al. 2018).

Error Across Time

Nonprobability sample accuracy is often expected to improve over time as online panels expand, recruitment methods mature, and post-stratification and weighting procedures become more sophisticated. The CES provides a strong test of this claim: across survey waves, it has repeatedly revised its post-stratification scheme and drawn respondents from multiple panel providers operating in a competitive marketplace.

Contrary to this expectation, CES accuracy did not improve over the 17 years studied. If anything, survey error trended modestly upward, though this trend was not statistically significant. In a variable fixed-effects model estimated on year-by-variable RMSE—including only variables that were secondary in all years and observed in every survey wave (Figure 4, Panel A)—RMSE increased by approximately 0.075 percentage points per year (HC3 robust SE = 0.045, $p = 0.092$; 95% CI: [-0.012, 0.163]). A permutation test that randomly reassigned year labels produced similar, marginal evidence against the null of no temporal structure ($p = 0.066$). Because the CES is fielded biennially, the point estimate corresponds to roughly 0.15 RMSE points per survey wave, or an estimated increase of approximately 1.2 RMSE points from 2006 to 2022.

Figure 4 Panel B disaggregates these trends by variable class and Table 11 reports model results. Demographic variables show a positive but only marginally significant increase in error over time (0.112 RMSE points per year; SE = 0.061; $p = 0.067$). The single voting administration variable, Voting Method, shows a larger and statistically significant upward trend (0.135; SE = 0.048; $p = 0.005$), though this estimate should be interpreted as a trend for that specific variable rather than for a broader class of voting administration measures. Candidate-choice variables show no meaningful temporal trend (0.006; SE = 0.092; $p = 0.949$).

Error as a Function of Respondent Count and Contest Competitiveness

In probability samples, increasing n reduces sampling variance at a well-defined $1/\sqrt{N}$ rate. In nonprobability samples such as the CES, a larger n still reduces random variability, but it does not necessarily reduce selection bias, coverage bias, bogus-response bias, or adjustment-model error (Elliott and Valliant 2017; Mercer, Kreuter, et al. 2017). This distinction matters because nonprobability samples are precisely the ones for which large n is cheap to obtain, which raises the question of whether a large nonprobability sample is preferable to a smaller probability sample of the same population.

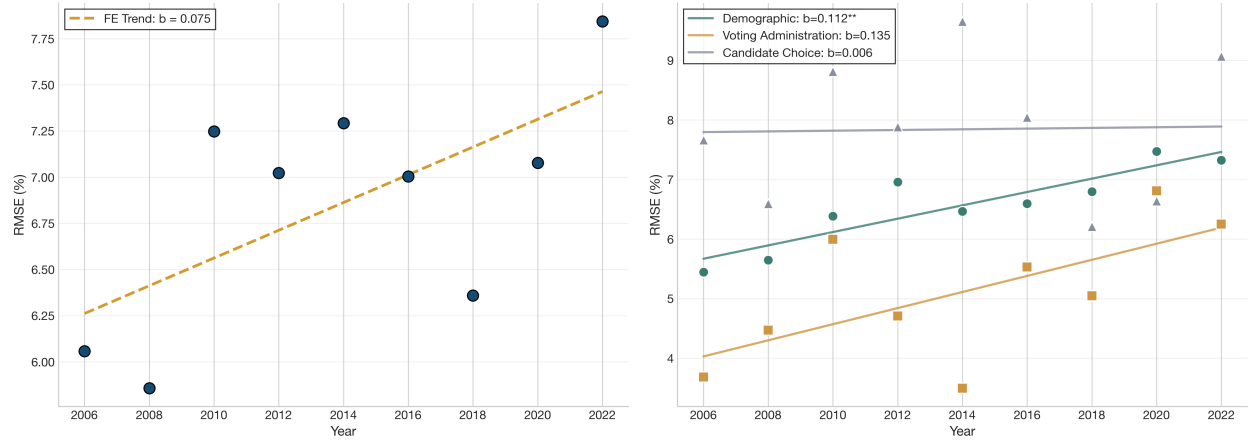


Figure 4: Secondary Variable RMSE Trends Over Time

RMSE trends over time for secondary variables that were observed in every survey wave and classified as secondary in all years. Eight variables are included: Employment Status, Family Income, Union Membership, Veteran Status, State Representative, State Senator, U.S. House, and Voting Method. **Panel A:** Each point represents the average RMSE across included variables in a given year. The dashed line shows the estimated linear trend from a variable fixed-effects regression estimated on year-by-variable RMSE, with HC3 heteroskedasticity-robust standard errors. The estimated coefficient indicates the average change in RMSE percentage points per calendar year, net of baseline differences across variables. **Panel B:** RMSE trends disaggregated by variable class. Reported coefficients come from the same fixed-effects specification estimated separately within each class, except for Voting Administration, which contains only one variable and therefore reduces to a year trend for Voting Method. Significance indicated by * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 11: Secondary Variable RMSE Trends Over Time

Model	β	SE	p -value	Perm. p
<i>Panel A: Fixed Effects Model</i>				
Variable FE	0.0751	0.0446	0.092	0.067
<i>Panel B: Simple OLS by Class</i>				
Demographic	0.1121**	0.0208	0.001	—
Voting Administration	0.1352	0.0597	0.058	—
Candidate Choice	0.0058	0.0826	0.946	—
Overall	0.0751	0.0340	0.063	—

Note: β represents the change in RMSE (percentage points) per year.

Panel A reports OLS with variable fixed effects, controlling for baseline differences between 8 consistently Secondary variables.

Panel B reports simple OLS regressions of mean RMSE on year, separately by variable class.

Robust standard errors (HC3) used for all models. Permutation p -value based on 10,000 permutations.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Meng’s “big data paradox” formalizes this concern. Using 2016 CES data, he shows that a self-reported vote-choice proportion from a sample of roughly 2.3 million respondents—about 1% of eligible voters—carries the same mean squared error as a genuine simple random sample of only $n \approx 400$, a reduction in effective sample size of about 99.98% (Meng 2018).

Comparing the respondent-count gradient in the CES to the $1/\sqrt{N}$ pattern from probability sampling indicates how much of the observed benchmark error was plausibly reducible through larger samples, as opposed to error that persisted even after sampling variability should have declined.

Table 12 reports variable-level models of benchmark error using two respondent-count specifications. The $\log(N)$ specification provides a flexible association test: it asks whether estimates based on larger respondent pools were closer to the benchmark, allowing the marginal benefit of additional respondents to diminish as sample size grows, since the difference between 25 and 100 respondents should matter more than the difference between 1,025 and 1,100 respondents. The $1/\sqrt{N}$ specification then imposes the functional form implied by probability-sample sampling error. If CES error followed ordinary sampling-error scaling, error should increase approximately linearly with $1/\sqrt{N}$. Together, the two specifications distinguish the general respondent-count association from the more specific probability-sampling benchmark.

The results show that respondent count matters, but unevenly. The clearest relationships appear for candidate-choice variables, especially offices whose estimates are built from small geographic or contest-specific respondent pools. U.S. House and State Representative estimates show statistically significant respondent-count gradients under both specifications, and several other candidate-choice offices have coefficients in the expected direction. Demographic variables, by contrast, show weak, inconsistent, and generally insignificant relationships with respondent count.

Figure 5 visualizes the same question, plotting observed RMSE across respondent-count bins and overlaying a fitted curve of the form

$$\text{RMSE}(N) = \alpha + \frac{\beta}{\sqrt{N}}.$$

Here α is the estimated error floor: because $\beta/\sqrt{N} \rightarrow 0$ as N grows, the curve approaches α at large respondent counts. The parameter β governs the sample-size-scaling component, with larger values implying that RMSE falls more steeply as respondent count increases. The curve is descriptive rather than structural: it asks whether the empirical pattern resembles the conventional sampling-error relationship while allowing error to flatten at a nonzero floor. Accordingly, α should be read as an empirical residual-error floor, not a theoretical lower

Table 12: Respondent Count and Absolute Error by Variable

Variable	$\log(N)$	$1/\sqrt{N}$	N	States	R^2
<i>Demographic</i>					
Age Group	0.191 (1.486)	21.694 (18.733)	459	51	0.584 / 0.586
Sex	0.116 (2.366)	46.248 (31.860)	459	51	0.311 / 0.324
Hispanic Origin	-1.062 (0.900)	26.991 (24.100)	357	51	0.947 / 0.947
Education	-0.023 (1.776)	-5.023 (33.895)	459	51	0.639 / 0.639
Family Income	-0.768 (0.889)	5.471 (13.300)	459	51	0.343 / 0.342
Employment Status	-2.232 (2.548)	69.139 (43.679)	459	51	0.595 / 0.604
Union Membership	-0.599 (1.156)	29.016 (19.896)	459	51	0.587 / 0.592
Veteran Status	-0.594 (1.287)	-19.965 (13.325)	459	51	0.656 / 0.658
Residence Duration	0.019 (1.840)	49.699 (49.390)	357	51	0.482 / 0.486
<i>Voting Administration</i>					
Voting Method	-0.298 (1.081)	-12.869 (28.643)	458	51	0.440 / 0.441
<i>Candidate Choice</i>					
President	-6.544** (2.192)	65.039 (50.070)	204	51	0.573 / 0.547
U.S. Senate	-3.345* (1.458)	69.228 (49.633)	303	50	0.503 / 0.499
U.S. House	-2.284*** (0.410)	38.470*** (7.973)	2,888	50	0.135 / 0.138
Governor	-3.877 (2.239)	104.378 (69.975)	225	45	0.526 / 0.526
Attorney General	-9.839** (3.767)	285.461** (88.939)	175	40	0.407 / 0.410
Secretary of State	2.036 (3.081)	-88.700 (109.474)	146	32	0.341 / 0.344
State Senator	0.412 (1.103)	-5.437 (17.411)	367	45	0.361 / 0.361
State Representative	-2.288** (0.863)	35.166*** (10.272)	397	46	0.509 / 0.508

Note: Entries report OLS coefficients with state-clustered standard errors in parentheses. The dependent variable is absolute percentage-point error. Each variable is modeled separately with state and year fixed effects. The $\log(N)$ column tests whether larger respondent counts are associated with lower error; the $1/\sqrt{N}$ column tests whether error scales with the theoretical sampling-error rate. Voting turnout is excluded. Estimates use unweighted CES estimates. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

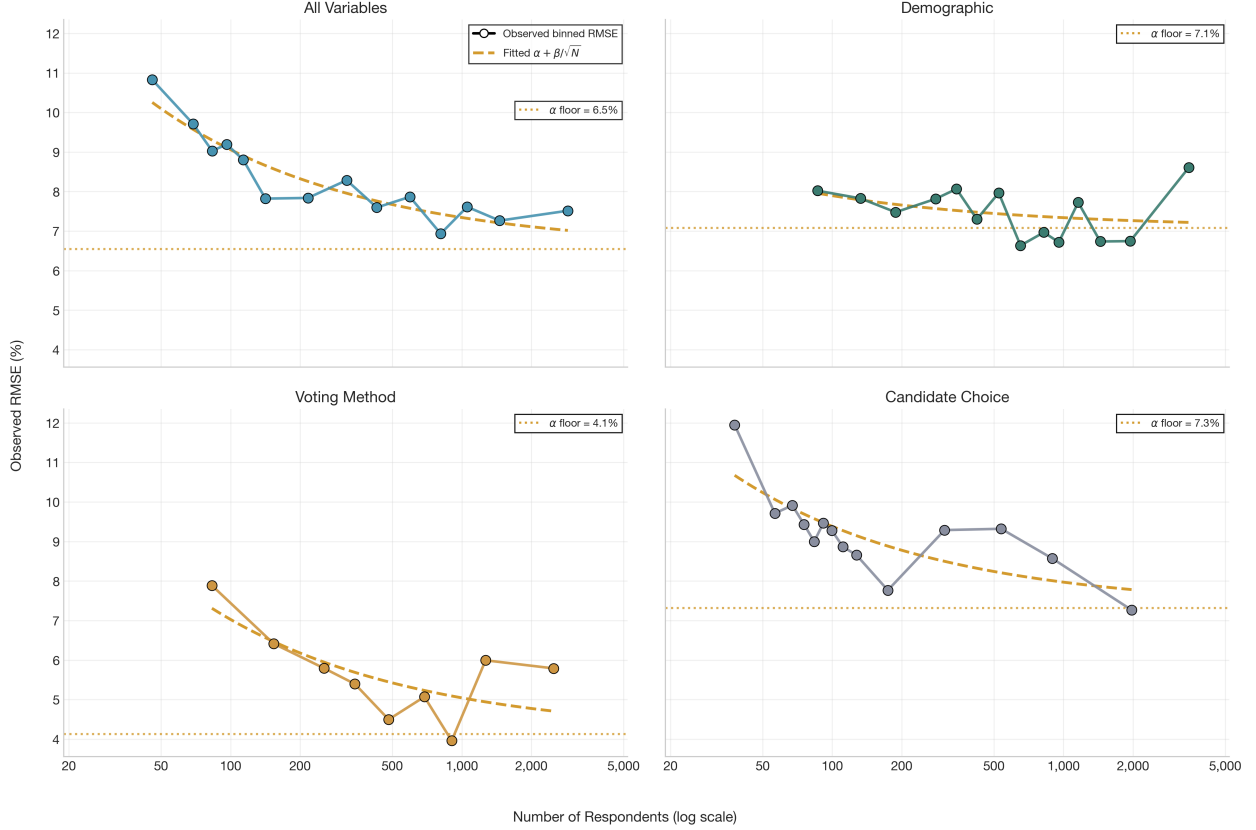


Figure 5: Observed RMSE by Respondent Count and $1/\sqrt{N}$ Fit

Points show observed RMSE across respondent-count bins. Dashed lines show fitted curves of the form $\alpha + \beta/\sqrt{N}$; dotted horizontal lines show the fitted α floor. Respondent count is shown on a log scale. Results use unweighted CES estimates and exclude voting turnout.

bound.

In the case of the CES, sample size does not appear meaningless. A larger respondent count is associated with lower error, but the relationship is noisy and bounded. Across all variables, RMSE declines as respondent count increases and mirrors the $1/\sqrt{N}$ shape expected under probability sampling, but the relationship flattens well above zero, with a fitted floor of approximately 6.6 percentage points. Candidate-choice variables show the clearest decline, consistent with the variable-level models, yet their fitted curve also levels off—around 7.4 percentage points. Voting-method items have a lower fitted floor, though the pattern is noisy given there is just one variable in that class. Demographic variables show little systematic improvement, with observed RMSE fluctuating around a nearly flat curve.

The analysis so far pools respondents within state-years. Because U.S. House estimates are constructed at the level of the congressional district rather than the state, they let us

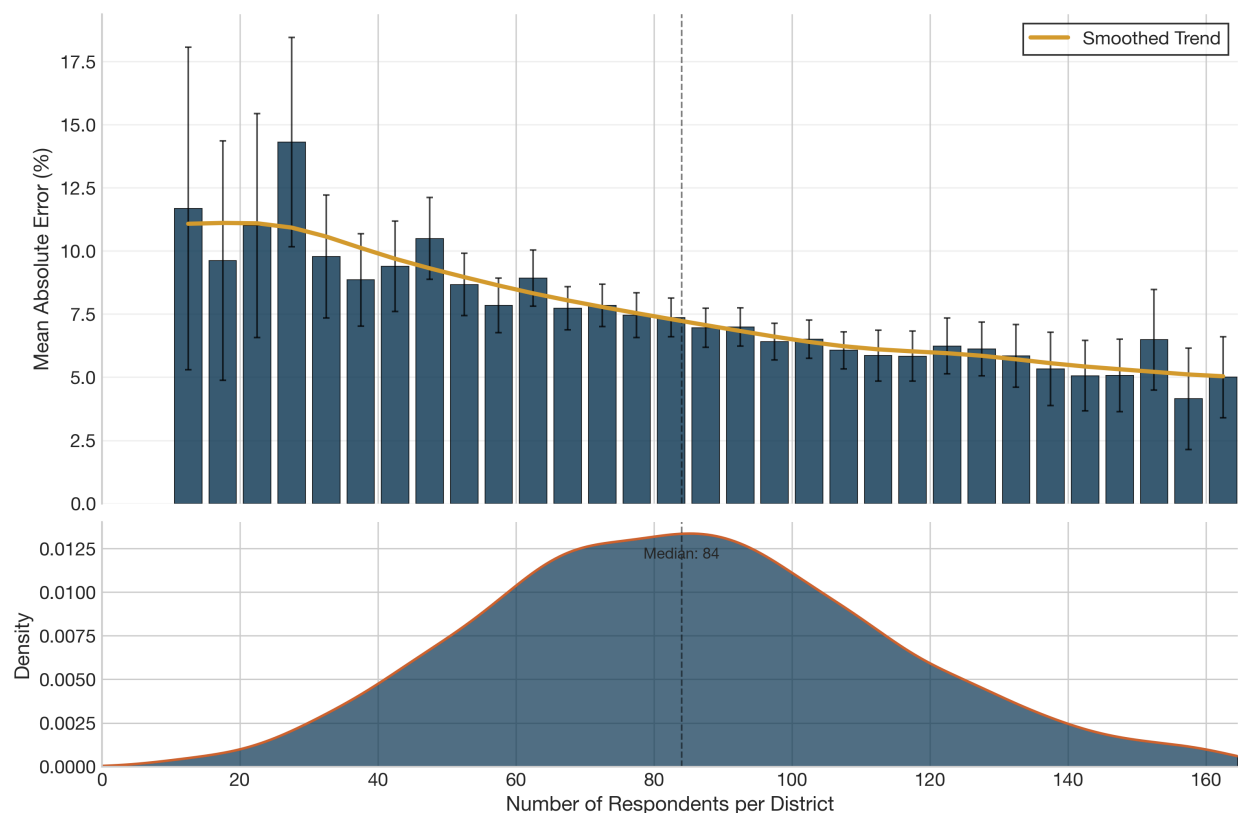


Figure 6: U.S. House Mean Absolute Error and Respondent Density by District

check whether the same respondent-count relationship holds at a finer level of aggregation. Figure 6 plots district-level U.S. House mean absolute error against per-district respondent count, alongside the distribution of respondents across districts. The district-level pattern mirrors the state-level one: error declines as the number of respondents per district rises.

Contest competitiveness introduces a separate source of heterogeneity, and it operates in the opposite direction from the naive expectation that close races are harder to measure. As the winning candidate’s benchmark vote share moves farther from 50%—that is, as races become less competitive—RMSE generally increases.

One possible driving mechanism is that in lopsided races the true vote split is far from parity, but CES respondents divide much closer to 50/50—partly because less salient contests invite weaker recall, and partly because turnout overreporting adds vote choices from people who did not vote. The result is a large gap between the survey-implied split and the benchmark winning share.

The increase is most pronounced in the least competitive races, including contests where a major-party candidate dropped out or only one major-party candidate appeared on the ballot. In these cases the benchmark winning share can approach or exceed 90%, while the

survey-implied Democratic/Republican split remains much closer to parity.

Table 13 and Figure 7 trace this pattern by progressively retaining races with higher winning vote shares. Moving from left to right includes less competitive contests. For most offices, RMSE is relatively stable through moderate cutoffs but rises once high winning-share races are included. The increase is especially visible for Attorney General, Secretary of State, and State Senator contests.

Table 13: Sensitivity of Office-Level RMSE to Less Competitive Races

Office	Maximum Winning Vote Share Retained				
	$\leq 60\%$	$\leq 70\%$	$\leq 80\%$	$\leq 90\%$	$\leq 100\%$
President	2.7 (120)	2.8 (175)	2.8 (178)	2.8 (178)	2.8 (182)
U.S. Senate	3.9 (146)	3.9 (239)	4.0 (262)	4.0 (262)	4.0 (264)
U.S. House	8.9 (1,016)	9.2 (2,039)	9.3 (2,517)	9.4 (2,675)	9.4 (2,823)
Governor	4.3 (129)	4.3 (176)	4.6 (188)	4.6 (188)	4.6 (188)
Attorney General	5.4 (95)	5.2 (143)	6.0 (154)	6.1 (156)	9.7 (167)
Secretary of State	5.4 (77)	5.7 (116)	6.2 (124)	6.9 (128)	9.6 (132)
State Senator	6.0 (204)	6.6 (298)	7.6 (337)	7.8 (345)	8.9 (348)
State Representative	5.3 (246)	5.9 (350)	6.3 (380)	6.3 (382)	6.3 (382)
Average	5.3	5.5	5.9	6.0	6.9

Note: Each cell reports RMSE in percentage points, with the number of races included in parentheses. Columns recalculate office-level RMSE after retaining only races where the winning candidate’s benchmark vote share is less than or equal to the listed cutoff. Moving from left to right progressively includes less competitive races. Estimates use matching + CES post-stratification.

U.S. House races let us examine these two forces jointly. Because the unit of inference is the district and many districts contain relatively few CES respondents, House contests combine competitiveness and small sample size in a way that single-office, state-level estimates do not. Appendix 4 presents a two-way RMSE matrix that stratifies U.S. House races by the winner’s vote share and by minimum district-level respondent thresholds, showing how—within states—competitiveness and district size interacted.

Comparison to Prior Evaluations of Error in the CES

Previous evaluations of the CES have reported accuracy metrics for selected offices and election years, though these analyses are more limited in scope and do not systematically distinguish between primary and secondary variables. Table 14 compares CES-reported RMSEs to RMSEs computed in this study.

RMSEs calculated in this study are consistently higher than CES-reported RMSEs for most offices. Differences are modest for presidential elections (+0.3 RMSE points on average) and U.S. Senate (-0.1), but larger for Attorney General (+4.0) and Secretary of State (+3.0).

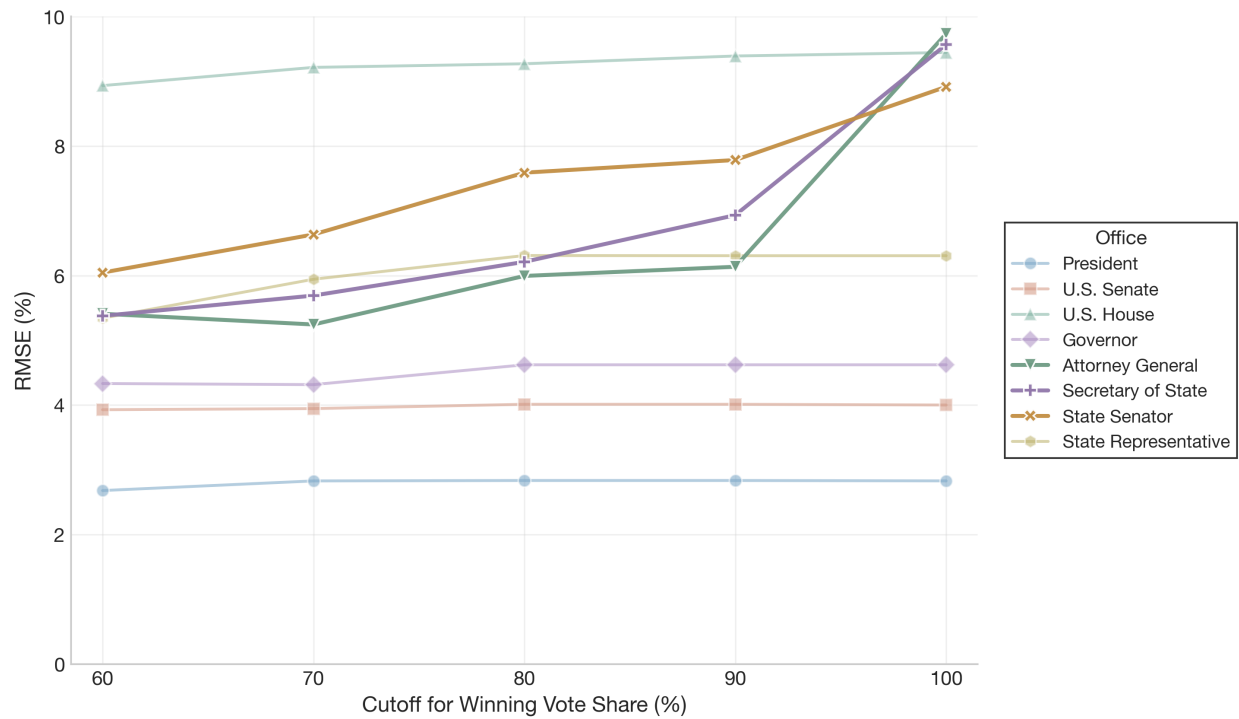


Figure 7: Sensitivity of Office-Level RMSE to Less Competitive Races

Each point recalculates office-level RMSE using only races where the winning candidate's benchmark vote share is at or below the x-axis cutoff. For example, 100% includes all contests for that office in the RMSE calculation, whereas 80% excludes races where the winning candidate received more than 80% of the vote. Moving from left to right progressively includes less competitive races. Error generally increases as less competitive contests are retained, with the sharpest increases occurring after very high winning-share races are included.

Table 14: CES-Reported RMSEs vs. RMSEs Calculated in This Study

Year	President			Governor			U.S. Senate			Attorney General			Secretary of State			Year Mean		
	Study	CES	Δ	Study	CES	Δ	Study	CES	Δ	Study	CES	Δ	Study	CES	Δ	Study	CES	Δ
2006	-	-	-	5.0	2.2	+2.7	5.9	4.3	+1.6	13.3	4.3	+9.0	11.2	4.4	+6.8	8.8	3.8	+5.0
2008	3.5	2.4	+1.1	4.7	3.2	+1.4	5.1	-	-	-	-	-	-	-	-	4.4	2.8	+1.3
2010	-	-	-	3.5	2.2	+1.2	2.0	1.3	+0.7	11.8	5.0	+6.8	6.4	4.5	+1.9	5.9	3.3	+2.7
2012	2.9	3.3	-0.4	3.4	-	-	2.3	3.7	-1.3	15.1	16.9	-1.7	10.9	9.9	+1.0	6.9	8.4	-0.6
2014	-	-	-	3.5	4.0	-0.4	3.1	3.4	-0.3	8.0	-	-	8.3	-	-	5.7	3.7	-0.4
2016	1.7	1.6	+0.1	4.0	4.0	-0.1	4.0	7.4	-3.4	4.9	-	-	9.6	-	-	4.9	4.3	-1.1
2018	-	-	-	4.5	4.1	+0.4	4.7	4.9	-0.2	4.8	-	-	4.2	3.4	+0.8	4.5	4.1	+0.3
2020	2.4	-	-	4.3	1.9	+2.4	3.3	-	-	3.5	1.9	+1.6	3.6	5.3	-1.7	3.4	3.1	+0.8
2022	-	-	-	5.7	-	-	4.8	2.4	+2.4	7.6	3.1	+4.4	13.9	4.4	+9.5	8.0	3.3	+5.4
Mean	2.6	2.4	+0.3	4.3	3.1	+1.1	3.9	3.9	-0.1	8.6	6.3	+4.0	8.5	5.3	+3.0	5.9	4.1	+1.5

- indicates data not available for that year-office combination.

Note: All RMSEs are computed using CES-provided post-stratification weights. “Study” shows RMSEs calculated in this analysis; “CES” shows RMSEs reported by the CES team. $\Delta = \text{Study} - \text{CES}$; positive values indicate this study calculated higher error than CES reported.

Source: CES-reported RMSEs from publicly-available CES data guides published alongside each year’s primary dataset (<https://cces.gov.harvard.edu/>).

Across years, calculated RMSEs exceed CES-reported values by an average of 1.5 RMSE points.

Matching-Only Error

Unweighted CES error reflects sample matching performance alone.

As shown in Table 15, average matching-only RMSE was 9.4% across years, and 8.1% when turnout was excluded, indicating that matching alone did not yield consistently high accuracy. Turnout-excluded RMSE reached 10.1% in 2022.

Table 15: Matching-Only RMSE by Variable and Year

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022	Yearly Mean
<i>Demographic</i>										
Age Group	7.7	6.3	12.3	9.7	7.4	8.6	7.4	8.8	8.3	8.5
Sex	5.1	5.2	3.2	3.6	4.0	6.1	5.3	6.3	5.7	5.0
Hispanic Origin	-	-	7.4	8.9	8.3	8.3	9.2	9.3	5.2	8.1
Education	5.5	3.9	13.8	7.8	5.0	7.4	6.5	6.2	7.1	7.0
Family Income	4.3	3.8	5.2	3.3	3.2	4.0	4.0	4.9	7.1	4.4
Employment Status	10.9	7.2	10.6	15.7	12.5	10.7	12.4	12.4	11.8	11.6
Union Membership	5.5	3.7	5.4	5.3	5.1	4.9	5.0	5.5	4.8	5.0
Veteran Status	8.6	7.2	11.0	9.3	6.8	5.4	5.3	4.3	4.5	6.9
Residence Duration	-	4.2	10.2	8.6	5.5	5.7	-	4.8	5.1	6.3
Demographic Avg.	6.8	5.2	8.8	8.0	6.4	6.8	6.9	7.0	6.6	6.9
<i>Voting Administration</i>										
Voting Turnout	47.3	30.1	43.7	37.7	43.3	35.3	40.2	30.7	40.1	38.7
Voting Method	4.3	4.0	4.7	6.2	3.8	5.3	5.0	8.5	8.0	5.5
Voting Administration Avg.	25.8	17.0	24.2	21.9	23.6	20.3	22.6	19.6	24.0	22.1
<i>Candidate Choice</i>										
President	-	5.8	-	4.4	-	8.1	-	10.0	-	7.1
U.S. Senate	5.6	8.0	6.1	4.3	5.8	7.3	6.3	9.9	9.9	7.0
U.S. House	4.3	10.6	9.3	7.7	8.5	8.0	9.4	10.1	11.4	8.8
Governor	4.9	4.8	6.0	2.8	4.8	7.1	6.5	10.9	10.9	6.5
Attorney General	12.3	-	12.6	15.4	9.3	6.5	7.8	11.3	12.8	11.0
Secretary of State	11.5	-	6.7	9.7	9.1	10.0	7.9	11.1	18.2	10.5
State Senator	10.1	5.9	9.7	8.0	15.4	9.4	7.5	13.4	14.6	10.5
State Representative	7.1	5.4	6.2	5.7	8.8	8.6	7.2	11.7	13.6	8.3
Candidate Choice Avg.	8.0	6.7	8.1	7.3	8.8	8.1	7.5	11.1	13.1	8.7
Yearly RMSE	9.7	7.2	10.2	9.2	9.3	8.8	9.0	10.0	11.1	9.4
Yearly RMSE (excl. Turnout)	7.7	6.0	8.8	8.4	7.9	7.5	7.3	9.3	10.1	8.1

- indicates CES did not measure vote choice for that office that year.

Table 16 compares matching-only to matching + post-stratification estimates, showing that weighting improved accuracy for variables included in the weighting scheme: primary variables exhibited RMSE that was 3.4 percentage points higher without weights. By contrast, post-stratification improved accuracy for secondary variables by only 1.0 points on

average. In years when they were weighted, turnout and presidential vote choice showed especially large improvements, with matching-only RMSE higher by 5.1 and 7.7 points respectively.

The contribution of post-stratification changed over time. Figure 8 shows that, averaged across all variables, the accuracy gains from weighting were relatively stable across years. But for secondary variables the impact of weighting appeared to grow over time, suggesting that post-stratification was doing progressively more work to reduce error even for variables not directly constrained by the weighting scheme. Disaggregating by variable type, the error reduction associated with weighting declined modestly for demographic variables but increased for candidate-choice outcomes. This pattern coincides with the inclusion of offices such as U.S. Senate and Governor in the post-stratification scheme beginning in 2014, which mechanically improves accuracy for those races.

Separating post-stratification effects by office, as in Figure 9, reveals a similar descriptive pattern: offices included in post-stratification generally exhibit lower overall error and larger apparent reductions in error attributable to weighting.

anesrake Weights

Finally, we evaluated how closely standard best-practice post-stratification applied to the nonprobability sample approached the accuracy achieved by YouGov’s bespoke weighting scheme. We constructed a single national set of calibration weights using **anesrake**. To ensure a fair comparison, we evaluated accuracy only on variables that were used in neither YouGov’s post-stratification nor anesrake weighting.

As a validation of the anesrake procedure, we assessed accuracy on the variables used to construct the calibration weights. Averaged across years, RMSE on these variables was 0.0% (Appendix 4), indicating that the calibration successfully aligned the weighted sample with national benchmarks. However, this accuracy did not extend to variables excluded from both weighting schemes—State Representative, State Senator, and U.S. House races (Figure 10). For these outcomes, anesrake generally performed worse than both matching alone and matching plus CES post-stratification, with the partial exception of 2020 and 2022, where it improved on matching alone but still underperformed YouGov’s weights. As reported in Table 17, while YouGov’s weights reduced RMSE by an average of 1.3% across the three variables not used for weighting, anesrake weights increased error by 1.2%.

Thus, while traditional post-stratification effectively reduced error for variables directly included in the weighting scheme, it did not generally improve accuracy for other outcomes in the nonprobability matched sample, replicating prior findings that best-practice demographic

Table 16: Difference in RMSE Between Matching-Only and Matching + Post-Stratification

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022	Yearly Mean
Primary Variables – Used to Construct CES Weights										
<i>Demographic</i>										
Age Group	-	+1.8	+7.0	+6.2	+3.3	+5.7	+4.0	+5.4	+5.3	+4.8
Sex	-	-1.5	-1.7	+2.1	+1.7	+5.1	+2.4	+3.7	+2.7	+1.8
Hispanic Origin	-	-	-	-	-	-	+0.5	+1.1	+1.9	+1.2
Education	-	+0.7	+7.4	+4.3	+1.2	+1.7	+2.3	+1.5	+3.5	+2.8
Demographic Avg.	-	+0.3	+4.3	+4.2	+2.1	+4.2	+2.3	+2.9	+3.3	+2.9
<i>Voting Administration</i>										
Voting Turnout	-	+4.2	+12.1	-1.0	-	-	-	-	-	+5.1
Voting Administration Avg.	-	+4.2	+12.1	-1.0	-	-	-	-	-	+5.1
<i>Candidate Choice</i>										
President	-	-	-	-	-	-	-	+7.7	-	+7.7
U.S. Senate	-	-	-	-	+2.8	+3.3	+1.6	+6.5	+5.1	+3.8
Governor	-	-	-	-	+1.2	+3.1	+2.1	+6.6	+5.1	+3.6
Candidate Choice Avg.	-	-	-	-	+2.0	+3.2	+1.8	+6.9	+5.1	+3.8
Yearly Delta	-	+1.3	+6.2	+2.9	+2.0	+3.8	+2.1	+4.7	+3.9	+3.4
Secondary Variables – Not Used to Construct CES Weights										
<i>Demographic</i>										
Age Group	+0.3	-	-	-	-	-	-	-	-	+0.3
Sex	-0.3	-	-	-	-	-	-	-	-	-0.3
Hispanic Origin	-	-	-0.3	+0.2	+0.3	-0.0	-	-	-	+0.0
Education	-7.5	-	-	-	-	-	-	-	-	-7.5
Family Income	-0.2	+0.3	+1.3	-0.4	-1.1	-0.4	-1.2	-1.8	-1.7	-0.6
Employment Status	+3.8	-3.6	-0.3	+1.8	+0.5	-2.1	-0.8	-1.4	-0.2	-0.2
Union Membership	+0.2	-0.0	-0.2	-0.5	+0.1	-0.3	-0.8	-0.7	-0.6	-0.3
Veteran Status	+3.8	+2.6	+5.9	+5.0	+2.3	+1.4	+2.3	+1.2	+1.3	+2.9
Residence Duration	-	-1.9	+2.8	+3.8	+0.4	+0.7	-	+0.6	-1.1	+0.8
Demographic Avg.	+0.0	-0.5	+1.5	+1.6	+0.4	-0.1	-0.1	-0.4	-0.5	+0.2
<i>Voting Administration</i>										
Voting Turnout	+3.5	-	-	-	-3.8	-1.3	+5.1	+4.3	-8.1	-0.1
Voting Method	+0.6	-0.5	-1.3	+1.4	+0.3	-0.3	-0.0	+1.7	+1.7	+0.4
Voting Administration Avg.	+2.0	-0.5	-1.3	+1.4	-1.8	-0.8	+2.5	+3.0	-3.2	+0.2
<i>Candidate Choice</i>										
President	-	+2.3	-	+1.5	-	+6.4	-	-	-	+3.4
U.S. Senate	-0.3	+2.9	+4.1	+2.0	-	-	-	-	-	+2.1
U.S. House	-0.7	+1.2	-2.1	-2.1	-0.5	-1.6	-0.1	+2.1	+1.5	-0.2
Governor	-0.0	+0.1	+2.5	-0.6	-	-	-	-	-	+0.5
Attorney General	-0.9	-	+0.7	+0.3	+1.3	+1.6	+3.0	+7.8	+5.3	+2.4
Secretary of State	+0.3	-	+0.3	-1.2	+0.8	+0.4	+3.7	+7.5	+4.3	+2.0
State Senator	-0.6	+0.4	+0.1	+0.6	+1.8	+1.4	+3.0	+7.0	+5.8	+2.2
State Representative	-0.3	+0.5	+0.8	-0.7	+2.4	+2.1	+2.6	+6.2	+5.1	+2.1
Candidate Choice Avg.	-0.4	+1.2	+0.9	-0.0	+1.2	+1.7	+2.4	+6.1	+4.4	+2.0
Yearly Delta	+0.1	+0.4	+1.0	+0.7	+0.4	+0.6	+1.5	+2.9	+1.1	+1.0

Note: Positive values indicate matching-only had greater error (post-stratification improved accuracy). CES 2006 documentation does not specify which variables were used for post-stratification, so all variables are assumed Secondary for that year.

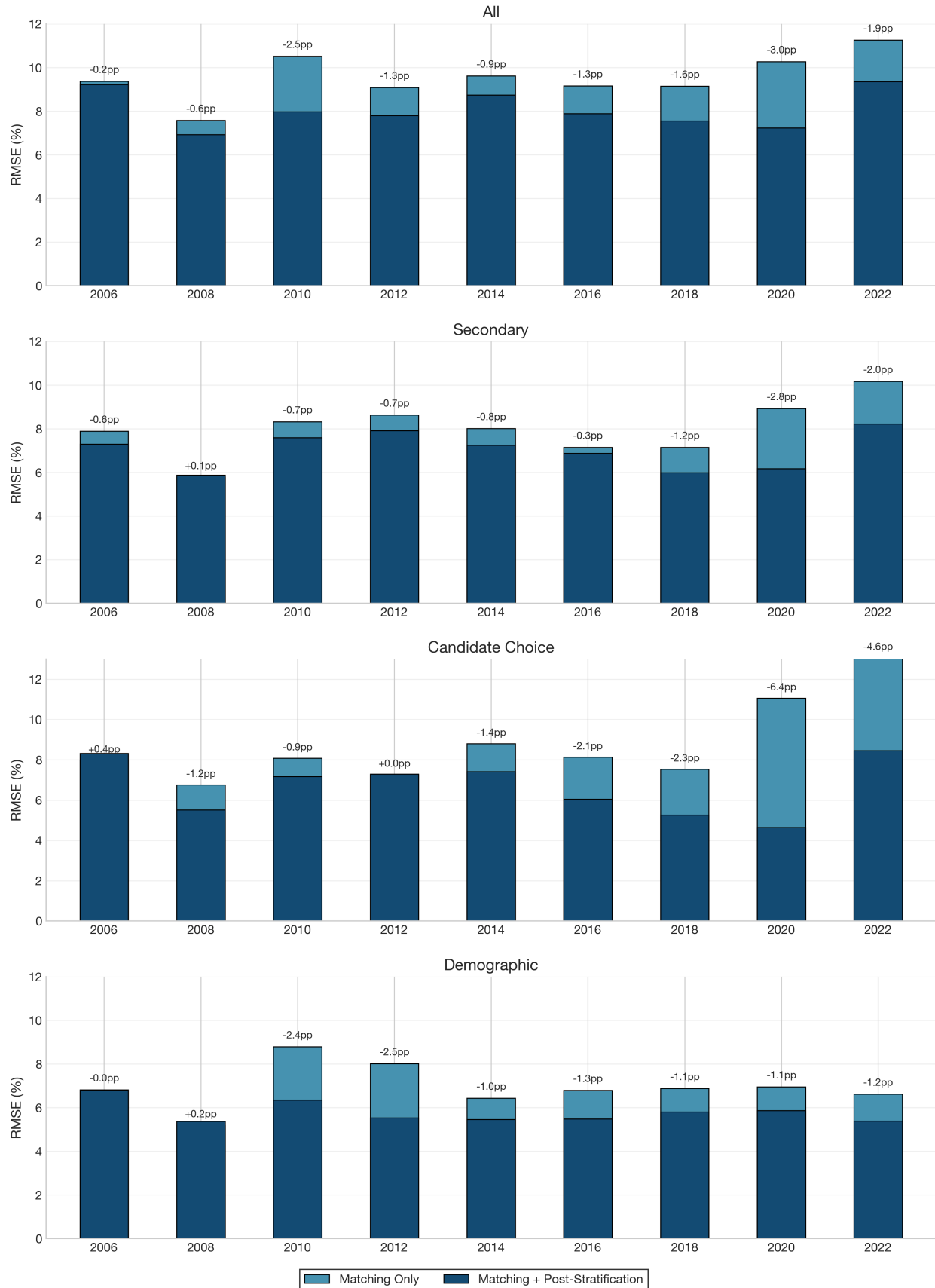


Figure 8: RMSE Reduction from CES Post-Stratification

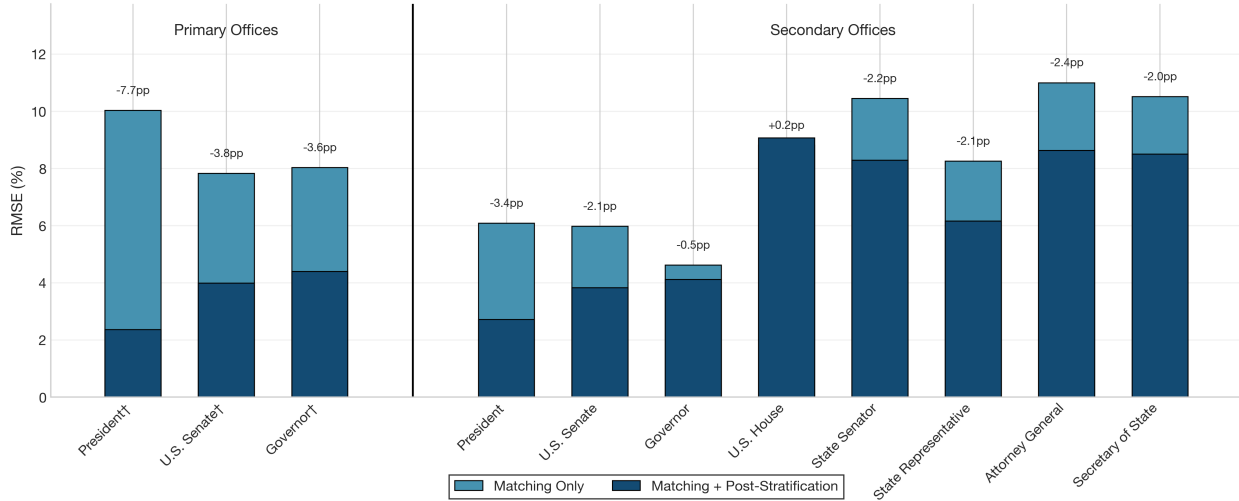


Figure 9: RMSE Reduction from CES Weighting by Office

weighting alone does not generalize to unconstrained variables in nonprobability surveys (Yeager et al. 2011; MacInnis et al. 2018).

Table 17: RMSE Changes by Weighting Method

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022	Mean
<i>Matching-Only RMSE (%)</i>										
State Representative	7.1	5.4	6.2	5.7	8.8	8.6	7.2	11.7	13.6	8.3
State Senator	10.1	5.9	9.7	8.0	15.4	9.4	7.5	13.4	14.6	10.5
U.S. House	4.3	10.6	9.3	7.7	8.5	8.0	9.4	10.1	11.4	8.8
Average	7.1	7.3	8.4	7.1	10.9	8.7	8.1	11.7	13.2	9.2
<i>Δ to Matching + Post-Strat. (pp)</i>										
State Representative	+0.3	-0.5	-0.8	+0.7	-2.4	-2.1	-2.6	-6.2	-5.1	-2.1
State Senator	+0.6	-0.4	-0.1	-0.6	-1.8	-1.4	-3.0	-7.0	-5.8	-2.2
U.S. House	+0.7	-1.2	+2.1	+2.1	+0.5	+1.6	+0.1	-2.1	-1.5	+0.2
Average	+0.5	-0.7	+0.4	+0.7	-1.2	-0.6	-1.8	-5.1	-4.2	-1.3
<i>Δ to ANESRake (pp)</i>										
State Representative	+3.0	+1.4	+1.3	+2.9	-1.2	+0.0	+1.7	-2.8	-0.4	+0.7
State Senator	+2.1	+1.0	+1.8	+2.6	-0.5	-0.2	+0.6	-3.1	-0.3	+0.4
U.S. House	+2.3	+1.8	+3.7	+3.8	+3.2	+1.6	+2.9	+2.5	+0.2	+2.5
Average	+2.5	+1.4	+2.3	+3.1	+0.5	+0.5	+1.7	-1.2	-0.1	+1.2

Note: Deltas show change in RMSE percentage points (pp) from Matching-Only. The three variables shown were the only variables present in all years and not used in either YouGov post-stratification or ANESRake.

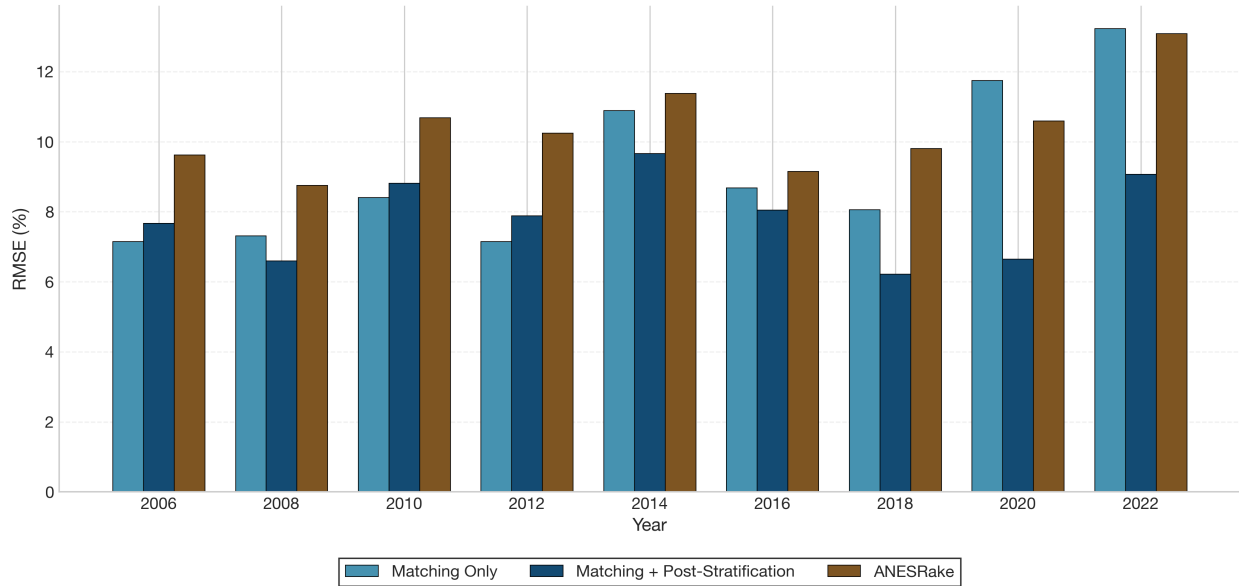


Figure 10: Comparison of Weighting Techniques Across Years

Includes U.S. House, State House, and State Senate—the only variables observed in all years and not used in either YouGov or anesrake weighting.

Discussion

This study provides the most comprehensive evaluation to date of the accuracy of matched-sample surveys using YouGov data from the Cooperative Election Study. Across nine survey waves spanning 2006–2022, CES estimates exhibited an average RMSE of approximately 8 percent, with errors reaching 10–15 percent in some years and for some variables. We find no evidence that CES accuracy has systematically improved over time, despite substantial changes to YouGov’s matching, weighting, and panel recruitment practices. In substantive terms, CES accuracy appears broadly comparable to—and in some cases worse than—what has been documented for other nonprobability survey designs.

A central finding is that accuracy gains in the CES are increasingly driven by YouGov’s bespoke post-stratification weights rather than by sample matching alone. When CES data are analyzed without weights, error increases meaningfully, indicating that matching by itself does not reliably approximate a probability sample. Conversely, applying YouGov’s proprietary weights reduces error, particularly for variables directly incorporated into the post-stratification scheme. In contrast, best-practice calibration weighting using anesrake—while effective at aligning the sample to national benchmarks on the variables it constrains—does not systematically improve accuracy for outcomes outside the weighting set. This result replicates a broader pattern observed in prior work: traditional demographic post-stratification

does not generalize well to unconstrained variables in nonprobability samples.

These findings align with, and in some respects extend, a growing literature documenting the limits of nonprobability survey accuracy. Prior studies have shown that average error in nonprobability samples is often only modestly worse than in probability samples, but that nonprobability designs tend to exhibit substantially greater variance, including a nontrivial share of very large errors. Our results are consistent with this pattern. Even when average RMSE appears moderate, the distribution of errors is highly skewed, with sizable deviations in particular years, offices, and states. In this sense, CES performance looks no better—and at times worse—than what has been reported for other nonprobability samples using simpler recruitment or weighting strategies. While YouGov’s approach is often described as exceptional, our results suggest that it remains subject to similar limitations to those other nonprobability sampling methods face.

Nonprobability sampling is now a permanent feature of the survey research landscape. YouGov, in particular, occupies a dominant position in academic and applied research, and its use of sample matching has been widely interpreted as a methodological breakthrough. Our evidence challenges that interpretation. Sample matching, as implemented in the CES, does not appear to deliver accuracy gains that distinguish it meaningfully from other nonprobability methods. Instead, accuracy improvements depend heavily on post-stratification choices that incorporate outcome-adjacent information—an approach that may be infeasible for many research applications that lack benchmarks. For those cases, our secondary-variable results are the more relevant guide to expected accuracy.

At the same time, our findings help reconcile apparent contradictions in the literature. Some prior evaluations of the CES report lower error in selected years or offices. We reproduce lower error estimates in some cases as well. However, we show that these conclusions are sensitive to which variables are evaluated and whether those variables were used in post-stratification. When considering the most informative outcomes—secondary variables that were not weighted and that appear consistently across years—accuracy remains persistently low and no evidence of improvement over time. This distinction between primary and secondary variables is crucial for interpreting CES performance and has been largely absent from prior evaluations.

The first of several limitations of this study is that because YouGov’s weighting scheme has evolved over time, only a limited set of variables can be classified as consistently secondary across all survey waves, many of which involve candidate choice. Measurement of these outcomes is itself imperfect: special elections, fusion voting, and changes in ballot structure sometimes make it difficult to determine precisely which contest a respondent had in mind. These sources of noise likely inflate error estimates to some degree. That said,

such limitations reflect real-world constraints faced by researchers using CES data and do not undermine the broader patterns we observe.

Our results also point toward important directions for future research. First, the high variance of nonprobability samples implies that the single-year evaluations still common in the literature can be misleading. Multi-year designs are essential for assessing stability, trends, and tail risk in survey accuracy. Second, the sharp contrast between the effectiveness of weighting at the national level and its breakdown at the state level warrants closer investigation. Understanding why calibration succeeds for national margins but fails to generalize across subnational units may shed light on the underlying sources of error in nonprobability samples. Finally, comparative evaluations across multiple vendors remain rare. The Pew study remains one of the few head-to-head assessments of YouGov relative to other providers; additional work of this kind would be valuable.

In sum, our findings challenge the narrative of sample matching as a final solution to the accuracy problems of nonprobability surveys. While matched samples combined with bespoke post-stratification can improve performance on selected variables, they do not escape the fundamental constraints that characterize nonprobability designs: substantial variance, sensitivity to weighting choices, and limited generalizability to unconstrained outcomes. Researchers relying on CES data should therefore interpret results with caution, particularly when drawing inferences about variables not explicitly targeted by the weighting process.

References

- Ansolabehere, Stephen and Douglas Rivers (2013). “Cooperative survey research”. *Annual Review of Political Science* 16.1, pp. 307–329.
- Ansolabehere, Stephen and Brian Schaffner (2014). “Does survey mode still matter? Findings from a 2010 multi-mode comparison”. *Political Analysis* 22.3, pp. 285–303.
- Baker, Reginald, J. Michael Brick, Nancy A. Bates, Mike Battaglia, Mick P. Couper, Jill A. Dever, Krista J. Gile, and Roger Tourangeau (2013). *Report of the AAPOR Task Force on Non-Probability Sampling*. Task Force. American Association for Public Opinion Research. URL: https://aapor.org/wp-content/uploads/2022/11/NPS_TF_Report_Final_7_revised_FNL_6_22_13-2.pdf.
- Berent, Matthew K, Jon A. Krosnick, and Arthur Lupia (2016). “Measuring Voter Registration and Turnout in Surveys: Do Official Government Records Yield More Accurate Assessments?” *Public Opinion Quarterly* 80.3, pp. 597–621. ISSN: 0033-362X. DOI: [10.1093/poq/nfw021](https://doi.org/10.1093/poq/nfw021). URL: <https://doi.org/10.1093/poq/nfw021> (visited on 01/16/2026).

-
- Bethlehem, Jelke (2010). "Selection Bias in Web Surveys". *International Statistical Review* 78.2. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1751-5823.2010.00112.x>, pp. 161–188. DOI: <https://doi.org/10.1111/j.1751-5823.2010.00112.x>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1751-5823.2010.00112.x>.
- Bradshaw, Tim (2008). "YouGov abandons plans to form hedge fund". *Financial Times*. URL: <https://www.ft.com/content/449f5fcc-9972-11dd-9d48-000077b07658> (visited on 11/23/2025).
- Brick, J. Michael, Jon Cohen, Sarah Cho, Scott Keeter, Kyley McGeeney, and Nancy Mathiowetz (2015). "Weighting and Sample Matching Effects for an Online Sample". *AAPOR Conference*. Hollywood, FL, USA.
- Buskirk, Trent and David Dutwin (2015). "Probability Samples—Meet Your Match! A Comparison of Two Distance Measures for Linking Nonprobability and Probability Based Samples". *International Total Survey Error Conference*. Baltimore, MD, USA.
- Chang, Linchi and Jon A. Krosnick (2009). "National surveys via RDD telephone interviewing versus the Internet: Comparing sample representativeness and response quality". *Public Opinion Quarterly* 73.4, pp. 641–678. DOI: [10.1093/poq/nfp075](https://doi.org/10.1093/poq/nfp075).
- Cochran, W. G. (1977). *Sampling Techniques*. 3rd ed. New York: John Wiley & Sons.
- Coleman, Stephen (2001). "Online Campaigning". *Parliamentary Affairs* 54.4, pp. 679–688. DOI: [10.1093/parlij/54.4.679](https://doi.org/10.1093/parlij/54.4.679). URL: <https://www.dhi.ac.uk/san/waysofbeing/data/citizenship-robson-coleman-2001a.pdf>.
- Cornesse, Carina, Annelies G Blom, David Dutwin, Jon A. Krosnick, Edith D De Leeuw, Stéphane Legleye, Josh Pasek, Darren Pennay, Benjamin Phillips, Joseph W Sakshaug, Bella Struminskaya, and Alexander Wenz (2020). "A Review of Conceptual Approaches and Empirical Evidence on Probability and Nonprobability Sample Survey Research". *Journal of Survey Statistics and Methodology* 8.1, pp. 4–36. ISSN: 2325-0984. DOI: [10.1093/jssam/snz041](https://doi.org/10.1093/jssam/snz041). URL: <https://doi.org/10.1093/jssam/snz041> (visited on 12/30/2024).
- Crampton, Thomas (2007). "About Online Surveys, Traditional Pollsters Are: (C) Somewhat Disappointed". *The New York Times*. URL: <https://www.nytimes.com/2007/05/31/business/media/31adco.html> (visited on 01/24/2025).
- Crossley, Archibald M (1937). "Straw polls in 1936". *Public Opinion Quarterly* 1.1, pp. 24–35.
- DeBell, Matthew and Jon A. Krosnick (2009). *Computing Weights for American National Election Study Survey Data*. ANES Technical Report series nes012427. Ann Arbor, MI, and Palo Alto, CA: American National Election Studies.

-
- DeBell, Matthew, Jon A. Krosnick, Katie Gera, David S. Yeager, and Michael P. McDonald (2020). "The Turnout Gap in Surveys: Explanations and Solutions". *Sociological Methods & Research* 49.4, pp. 1133–1162. ISSN: 0049-1241. DOI: [10.1177/0049124118769085](https://doi.org/10.1177/0049124118769085). (Visited on 01/16/2026).
- Dever, Jill A. (2018). "Combining Probability and Nonprobability Samples to form Efficient Hybrid Estimates: An Evaluation of the Common Support Assumption". *Proceedings of the Federal Committee on Statistical Methodology (FCSM) Research Conference*. Washington, DC. URL: https://nces.ed.gov/FCSM/pdf/A4_Dever_2018FCSM.pdf.
- DiSogra, Charles, Stacie Greby, K. P. Srinath, Andrew Burkey, Carla Black, John Sokolowski, Xan Yue, Sarah Ball, and Sara Donahue (2015). "Matching an Internet Panel Sample of Health Care Personnel to a Probability Sample". *AAPOR Conference*. Hollywood, FL, USA.
- Dutwin, David and Trent Buskirk (2017). "Apples to Oranges or Gala versus Golden Delicious?: Comparing Data Quality of Nonprobability Internet Samples to Low Response Rate Probability Samples". *Public Opinion Quarterly* 81 (S1), pp. 213–239. ISSN: 0033-362X. DOI: [10.1093/poq/nfw061](https://doi.org/10.1093/poq/nfw061). URL: <https://doi.org/10.1093/poq/nfw061>.
- Elliott, Michael R. and Richard Valliant (2017). "Inference for nonprobability samples". *Statistical Science* 32.2, pp. 249–264.
- Erens, Bob, Sarah Burkill, Mick P. Couper, Frederick Conrad, Soazig Clifton, Clare Tanton, Andrew Phelps, Jessica Datta, Catherine H. Mercer, Pam Sonnenberg, Philip Prah, Kirstin R. Mitchell, Kaye Wellings, Anne M. Johnson, and Andrew J. Copas (2014). "Nonprobability Web surveys to measure sexual behaviors and attitudes in the general population: a comparison with a probability sample interview survey". *Journal of Medical Internet Research* 16.12, e276. DOI: [10.2196/jmir.3382](https://doi.org/10.2196/jmir.3382).
- Flood, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, Daniel Backman, Annie Chen, Grace Cooper, Stephanie Richards, Megan Schouweiler, and Michael Westberry (2024). *IPUMS CPS: Version 12.0*. Version 12.0. DOI: <https://doi.org/10.18128/D030.V12.0>. URL: <https://cps.ipums.org/cps/>.
- Hainmueller, Jens (2012). "Entropy Balancing for Causal Effects: A Multivariate Reweighting Method to Produce Balanced Samples in Observational Studies". *Political Analysis* 20.1, pp. 25–46. ISSN: 1047-1987, 1476-4989. DOI: [10.1093/pan/mpr025](https://doi.org/10.1093/pan/mpr025). (Visited on 12/31/2024).
- Hopkins, Daniel J., William Halm, Melissa Huerta, and Josearmando Torres (2024). "Getting the Race Wrong: A Case Study of Sampling Bias and Black Voters in Online, Opt-In Polls". *The Journal of Race, Ethnicity, and Politics* 9.3, pp. 429–441. DOI: [10.1017/rep.2024.11](https://doi.org/10.1017/rep.2024.11).

-
- Hur, Aram and Christopher H. Achen (2013). “Coding Voter Turnout Responses in the Current Population Survey”. *Public Opinion Quarterly* 77.4, pp. 985–993. ISSN: 0033-362X. DOI: [10.1093/poq/nft042](https://doi.org/10.1093/poq/nft042). URL: <https://doi.org/10.1093/poq/nft042> (visited on 12/16/2024).
- Jaro, Matthew A. (1989). “Advances in Record-Linkage Methodology as Applied to Matching the 1985 Census of Tampa, Florida”. *Journal of the American Statistical Association* 84.406, pp. 414–420. DOI: [10.1080/01621459.1989.10478785](https://doi.org/10.1080/01621459.1989.10478785).
- Kellner, Peter (2004). “Can Online Polls Produce Accurate Findings?” *International Journal of Market Research* 46.1, pp. 3–22. DOI: [10.1177/147078530404600102](https://doi.org/10.1177/147078530404600102). URL: <https://doi.org/10.1177/147078530404600102>.
- Kennedy, Courtney, Andrew Mercer, Scott Keeter, Nick Hatley, Kyley McGeeney, and Alejandra Gimenez (2016). “Evaluating Online Nonprobability Surveys”. *Pew Research Center*. URL: <https://www.pewresearch.org/methods/2016/05/02/evaluating-online-nonprobability-surveys/>.
- Kullback, Solomon (1959). *Information Theory and Statistics*. New York: Wiley.
- Lavrakas, Paul John, Darren Pennay, Dina Neiger, and Benjamin Phillips (2022). “Comparing Probability-Based Surveys and Nonprobability Online Panel Surveys in Australia: A Total Survey Error Perspective”. *Survey Research Methods* 16.2, pp. 241–266. ISSN: 1864-3361. DOI: [10.18148/srm/2022.v16i2.7907](https://doi.org/10.18148/srm/2022.v16i2.7907). URL: <https://ojs.ub.uni-konstanz.de/srm/article/view/7907> (visited on 01/16/2026).
- MacInnis, Bo, Jon A. Krosnick, Annabell S Ho, and Mu-Jung Cho (2018). “The Accuracy of Measurements with Probability and Nonprobability Survey Samples: Replication and Extension”. *Public Opinion Quarterly* 82.4, pp. 707–744. ISSN: 0033-362X. DOI: [10.1093/poq/nfy038](https://doi.org/10.1093/poq/nfy038). URL: <https://doi.org/10.1093/poq/nfy038>.
- McDonald, Michael P. (2025). *Voter Turnout*. URL: <https://election.lab.ufl.edu/voter-turnout/> (visited on 01/26/2025).
- Meng, Xiao-Li (2018). “Statistical paradises and paradoxes in big data (I): Law of large populations, big data paradox, and the 2016 US presidential election”. *The Annals of Applied Statistics* 12.2. ISSN: 1932-6157. DOI: [10.1214/18-AOAS1161SF](https://projecteuclid.org/journals/annals-of-applied-statistics/volume-12/issue-2/Statistical-paradises-and-paradoxes-in-big-data-I--Law/10.1214/18-AOAS1161SF.full). URL: <https://projecteuclid.org/journals/annals-of-applied-statistics/volume-12/issue-2/Statistical-paradises-and-paradoxes-in-big-data-I--Law/10.1214/18-AOAS1161SF.full> (visited on 07/05/2026).
- Mercer, Andrew, Courtney Kennedy, and Scott Keeter (2024). *Online opt-in polls can produce misleading results, especially for young people and Hispanic adults*. Pew Research Center. URL: <https://www.pewresearch.org/short-reads/2024/03/05/online->

[opt-in-polls-can-produce-misleading-results-especially-for-young-people-and-hispanic-adults/](#).

- Mercer, Andrew, Frauke Kreuter, Scott Keeter, and Elizabeth Stuart (2017). “Theory and Practice in Nonprobability Surveys: Parallels between Causal Inference and Survey Inference”. *Public Opinion Quarterly* 81 (S1), pp. 250–271. ISSN: 0033-362X. DOI: [10.1093/poq/nfw060](#). URL: <https://doi.org/10.1093/poq/nfw060> (visited on 07/05/2026).
- Mercer, Andrew, Arnold Lau, and Courtney Kennedy (2018). *For weighting online opt-in samples, what matters most?* Pew Research Center. URL: <https://www.pewresearch.org/methods/2018/01/26/for-weighting-online-opt-in-samples-what-matters-most/>.
- MIT Election Data and Science Lab (2017). *U.S. President 1976–2020*. Version V8. Section: 2017-11-01 15:23:09.819. DOI: [10.7910/DVN/42MVDX](#). URL: <https://doi.org/10.7910/DVN/42MVDX>.
- Pasek, Josh (2018). *anesrake: ANES Raking Implementation*. URL: <https://CRAN.R-project.org/package=anesrake>.
- Rivers, Douglas (2006). *Sample Matching*. White paper. Polimetrix, pp. 1–8. URL: https://jwolf-ra.wdfiles.com/local--files/web-panels/Rivers_SampleMatching.pdf.
- (2007). “Sampling for web surveys”. *Joint Statistical Meetings*. Vol. 4. American Statistical Association Alexandria, VA. URL: https://static.texastribune.org/media/documents/Rivers_matching4.pdf (visited on 01/26/2025).
- Rivers, Douglas and Delia Bailey (2009). “Inference from Matched Samples in the 2008 U.S. National Elections”. *Proceedings of the Joint Statistical Meetings*. Washington, DC, USA. URL: <http://www.asasrms.org/Proceedings/y2009/Files/303309.pdf>.
- Stuart, Elizabeth A. (2010). “Matching Methods for Causal Inference: A Review and a Look Forward”. *Statistical Science* 25.1, pp. 1–21. ISSN: 0883-4237, 2168-8745. DOI: [10.1214/09-STS313](#). URL: <https://projecteuclid.org/journals/statistical-science/volume-25/issue-1/Matching-Methods-for-Causal-Inference--A-Review-and-a/10.1214/09-STS313.full> (visited on 12/31/2024).
- Sturgis, Patrick, Jouni Kuha, Nick Baker, Mario Callegaro, Stephen Fisher, Jane Green, Will Jennings, Benjamin E Lauderdale, and Patten Smith (2018). “An Assessment of the Causes of the Errors in the 2015 UK General Election Opinion Polls”. *Journal of the Royal Statistical Society Series A: Statistics in Society* 181.3, pp. 757–781.
- Szolnoki, Gergely and Dieter Hoffmann (2013). “Online, face-to-face and telephone surveys—Comparing different sampling methods in wine consumer research”. *Wine Economics and Policy* 2.2, pp. 57–66.

-
- Track Record: 2005 Errors* (2005). Electoral Calculus. URL: https://www.electoralcalculus.co.uk/trackrecord_05errors.html (visited on 11/23/2025).
- Vavreck, Lynn and Douglas Rivers (2008). “The 2006 Cooperative Congressional Election Study”. *Journal of Elections, Public Opinion and Parties*. DOI: [10.1080/17457280802305177](https://doi.org/10.1080/17457280802305177). URL: <https://www.tandfonline.com/doi/abs/10.1080/17457280802305177> (visited on 12/22/2024).
- Walker, Robert, Raymond Pettit, and Joel Rubinson (2009). “The Foundations of Quality Initiative: A Five-Part Immersion into the Quality of Online Research”. *Journal of Advertising Research* 49.4, pp. 464–485. ISSN: 0021-8499. DOI: [10.2501/S0021849909091089](https://doi.org/10.2501/S0021849909091089). URL: <https://doi.org/10.2501/S0021849909091089> (visited on 01/16/2026).
- Wang, Wei, David Rothschild, Sharad Goel, and Andrew Gelman (2015). “Forecasting elections with non-representative polls”. *International Journal of Forecasting* 31.3, pp. 980–991.
- Winkler, William E. (1990). *String Comparator Metrics and Enhanced Decision Rules in the Fellegi–Sunter Model of Record Linkage*. Research Report RR91/08. Washington, DC: U.S. Bureau of the Census.
- Wright, Kevin B. (2017). “Researching Internet-Based Populations: Advantages and Disadvantages of Online Survey Research, Online Questionnaire Authoring Software Packages, and Web Survey Services”. *Journal of Computer-Mediated Communication* 10.3, JCMC1034. ISSN: 1083-6101. DOI: [10.1111/j.1083-6101.2005.tb00259.x](https://doi.org/10.1111/j.1083-6101.2005.tb00259.x). URL: <https://doi.org/10.1111/j.1083-6101.2005.tb00259.x>.
- Yeager, David S., Jon A. Krosnick, LinChiat Chang, Harold S. Javitz, Matthew S. Levendusky, Alberto Simpser, and Rui Wang (2011). “Comparing the Accuracy of RDD Telephone Surveys and Internet Surveys Conducted with Probability and Non-Probability Samples”. *Public Opinion Quarterly* 75.4, pp. 709–747. ISSN: 0033-362X. DOI: [10.1093/poq/nfr020](https://doi.org/10.1093/poq/nfr020). URL: <https://doi.org/10.1093/poq/nfr020>.

Appendix

Table A1: National ANESRake-Weighted RMSE for Variables Used in Weighting

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022	Yearly Mean
ANESRake Full Variables										
<i>Demographic</i>										
Age Group	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sex	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Hispanic Origin	-	-	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0
Education	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Family Income	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Employment Status	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Union Membership	0.0	-	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Veteran Status	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Residence Duration	-	0.0	0.1	0.0	0.0	-	-	0.0	0.0	0.0
Demographic Avg.	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
<i>Voting Administration</i>										
Voting Turnout	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.2	0.0	0.1
Voting Method	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Voting Administration Avg.	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.1	0.0	0.0
Yearly RMSE (Full)	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ANESRake Restricted Variables										
<i>Demographic</i>										
Age Group	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Education	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Family Income	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Employment Status	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Demographic Avg.	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Yearly RMSE (Restricted)	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Note: RMSE in percentage points. National-level errors. ANESRake Full uses all available demographic and voting administration variables. ANESRake Restricted uses a subset (Education, Employment Status, Voter Registration, Voting Turnout).

Table A2: CPS Sample Description, 2006 – 2022

Year	Sample Size	Probability Sample?	Completion Rate	Field Dates
2006	1,221,832	Yes	91.7%	Monthly
2008	1,209,862	Yes	91.7%	Monthly
2010	1,230,780	Yes	92.0%	Monthly
2012	1,215,614	Yes	90.4%	Monthly
2014	1,218,742	Yes	88.9%	Monthly
2016	1,202,067	Yes	86.7%	Monthly
2018	1,150,355	Yes	84.4%	Monthly
2020	1,001,837	Yes	74.3%	Monthly
2022	964,927	Yes	72.9%	Monthly

Source: U.S. Bureau of Labor Statistics. https://www.bls.gov/cps/methods/response_rates.htm

Note: Sample sizes are annual totals across all monthly surveys. Completion rates are annual averages.

Table A3: U.S. House District-Level Accuracy by Sample Size and Race Competitiveness

Min. Respondents	Winner's Vote Share (%)				
	20–40	40–60	60–80	80–90	90–100
<i>RMSE (%)</i>					
All	10.0	8.9	9.5	11.1	10.4
30+	10.0	8.8	9.3	11.2	9.9
60+	4.3	8.2	8.7	10.3	10.6
90+	-	7.4	7.4	10.5	13.0
120+	-	6.8	6.7	8.9	20.3
<i>Number of District-Years</i>					
All	5	1076	1501	158	148
30+	5	1068	1469	151	141
60+	3	916	1221	105	85
90+	0	575	608	39	25
120+	0	216	184	8	2

Note: Each cell reports RMSE (in percentage points) averaged across all congressional district-years (2006–2022) meeting both the respondent threshold (row) and winner's vote share range (column). A district-year is a single congressional district in a single election year; the same district appears multiple times across the study period. Two patterns are notable. First, among competitive and moderately uncompetitive races, RMSE generally declines as the minimum respondent threshold increases, suggesting that district-level sample size contributes to House vote-choice error. Second, error remains elevated in very safe districts, and the respondent-count gradient weakens in those cells, suggesting that some of the error in lopsided races reflects factors other than finite-sample variability. The highest-threshold estimates in the 90–100% column should be interpreted cautiously because they are based on very small numbers of district-years.

Table A4: Data Sources

Category	Source	URL
Presidential	MIT Election Data and Science Lab	https://electionlab.mit.edu/research
U.S. Senate & House	Federal Election Commission	https://www.fec.gov/introduction-campaign-finance/election-results-and-voting-information/
Gubernatorial	CQ Press Elections Collection	https://library.cqpress.com/elections/
State legislative/executive	Ballotpedia	https://ballotpedia.org/Election_results
Demographics	IPUMS CPS (U.S. Census Bureau)	https://cps.ipums.org/cps/
Turnout	University of Florida Election Lab	https://election.lab.ufl.edu/research/voter-turnout/
Voting method	IPUMS CPS	https://cps.ipums.org/cps/

Note: Data cover federal election years 2006–2022.

Entropy Balancing Procedure

Beginning in 2014, the Cooperative Election Study (CES) weighted the matched sample to the target sample using entropy balancing (Hainmueller 2012). Entropy balancing is a multivariate reweighting method that assigns scalar weights to individual observations such that the reweighted sample exactly satisfies a set of covariate balance constraints while remaining as close as possible to a set of base weights.

Formally, weights were obtained by minimizing Kullback–Leibler divergence (Kullback 1959) between the final weights w_i and base weights q_i :

$$\min H(w) = \sum_i w_i \log\left(\frac{w_i}{q_i}\right),$$

subject to the following constraints:

$$\sum_{i \in S} w_i c_{ri}(X_i) = \mu_r, \quad (\text{A1})$$

$$\sum_{i \in S} w_i = 1, \quad (\text{A2})$$

$$w_i \geq 0 \quad \forall i. \quad (\text{A3})$$

Here, S indexes the matched sample, w_i denotes the post-stratification weight assigned to respondent i , and q_i denotes the base weight. In settings where base weights are unavailable—as is typical in nonprobability samples—base weights were set uniformly such that $q_i = 1/N$, where N is the matched sample size.

Constraint (A1) enforces balance on the r -th moment of covariate X in the target sample, where μ_r denotes the corresponding target-sample moment. In the CES implementation, moment functions were specified as

$$c_{ri}(X_{ij}) = X_{ij} - \mu_j,$$

where X_{ij} denotes the value of covariate j for respondent i , and μ_j is the target-sample mean of covariate j . Constraints (A2) and (A3) ensure that the resulting weights form a valid probability distribution.

Minimizing Kullback–Leibler divergence subject to these constraints yields a set of weights that deviate minimally from the base weights while achieving exact balance on the specified covariates. This property ensures that weights vary smoothly across observations and preserves their interpretation as probability weights.

Inconsistencies and Intricacies of Election Information Reporting

Candidate names occasionally differed in format between the CES questionnaire and official election returns. Jaro–Winkler fuzzy string matching (Jaro 1989; Winkler 1990) was used to map CES candidates to those in the election returns, and all matches were manually reviewed.

Some states did not report candidate party affiliation in official election returns, even when candidates were formally affiliated with a party. In these cases, candidates were manually assigned to their affiliated party to avoid artificially inflating CES error, as such candidates would otherwise have been treated as unaffiliated despite being identified by CES respondents as Democrats or Republicans.

In states such as New York and Vermont, candidates endorsed by multiple parties appeared multiple times on the ballot—once for each party line. For example, a candidate may have received votes under the Democratic Party, Working Families Party, and Progressive Party. In these cases, all votes in the election returns were aggregated to the candidate’s major-party affiliation (Democratic or Republican).

When a state held a special election on the same day as a general election, those votes were included. When both a special and a general election for the same office occurred on the same day, only the general election was counted.

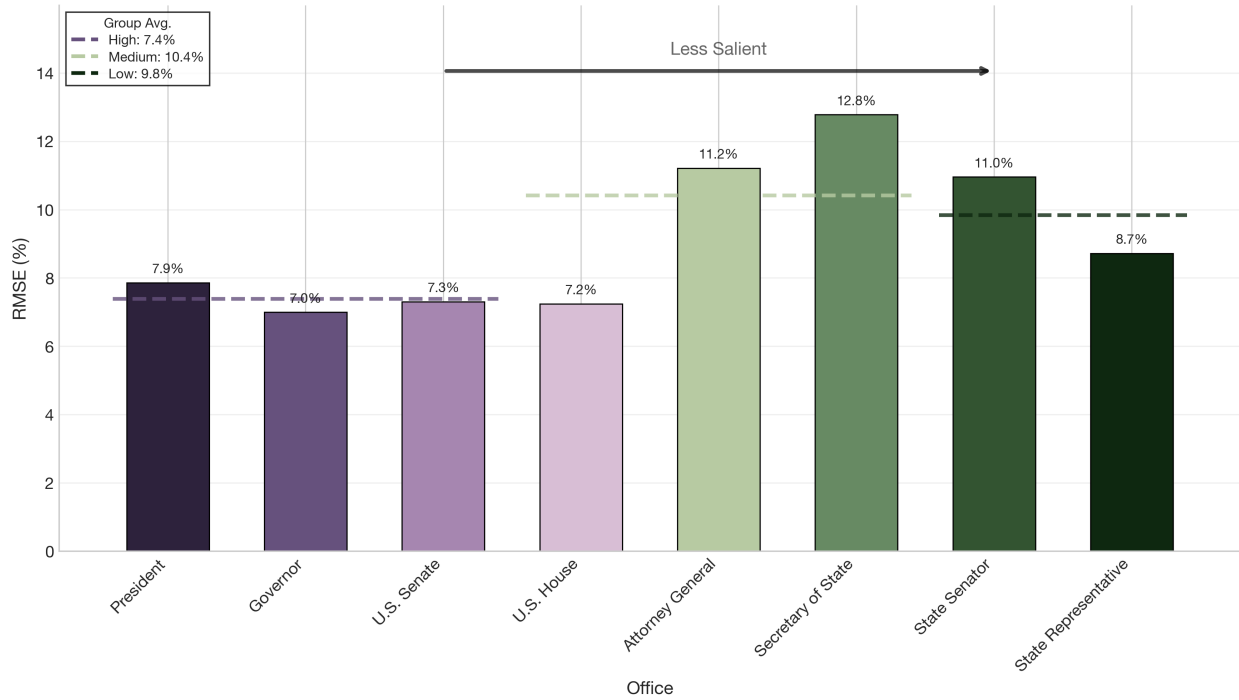


Figure A1: RMSE by Office Salience, Yearly Average

Race Salience

We report a supplementary analysis examining whether CES accuracy varies with race salience. Prior work in election polling suggests that higher-salience contests are typically measured more accurately than down-ballot races. Although the CES captures retrospective vote reports rather than pre-election intentions, it is useful to assess whether a similar pattern appears in these data.

To minimize potential confounds, the analysis used unweighted data measured at the party level. This specification avoids the influence of post-stratification weighting and ensures comparability across offices by holding the measurement scale constant.

Under this specification, RMSE was lowest for high-salience contests (7.4%). Error was higher for down-ballot offices, though the pattern was not strictly monotonic: medium-salience races had the highest average RMSE (10.4%), while low-salience offices averaged 9.8%. Figure A1 presents these results by salience category.